Project Verification Report

2021

COVER PAGE

Project Verification Report Form (VR)

BASIC INFORMATION	
Name of approved UCR Project Verifier / Reference No.	KBS Certification Services Limited https://www.ucarbonregistry.io/CouRegistry/VerifierList
Type of Accreditation	☒ CDM or other GHG Accreditation ☐ ISO 14065 Accreditation Name of the entity that provided the accreditation: UNFCCC Date of validity: 08/11/2024 to 09/11/2029 Web link of the active accreditation certificate and approval: https://cdm.unfccc.int/DOE/list/DOE.html?entityCode=E-0051
Approved UCR Scopes and GHG Sectoral scopes for Project Verification	SS 1: Energy industries (renewable/non-renewable sources) SS 13: Waste handling and disposal
Validity of UCR approval of Verifier	15/01/2022 onwards
Completion date of this VR	18/05/2026
Title of the project activity	Integrated Municipal Solid Waste Management Activities Programme, Nashik, India
Project reference no. (as provided by UCR Program)	492

Name of Entity requesting verification service (can be Project Owners themselves or any Entity having authorization of Project Owners, example aggregator.)	Nashik Waste Management Pvt. Ltd. (NWMPL)
Contact details of the representative of the Entity, requesting verification service (Focal Point assigned for all communications)	Dr. Priti Mastakar Independent Consultant – Financial, Economic and Environmental Modelling, Sustainability & Carbon Accounting Telephone: +91-7559143174 E-Mail ID: pritibalamastakar@gmail.com
Country where project is located	India
Applied methodologies (approved methodologies by UCR Standard used)	ACM0022: Large-scale: Consolidated Methodology: Alternative waste treatment processes, Version 03.0 AMS-I. D: Small-scale Methodology: Grid connected renewable electricity generation, Version 18.0
GHG Sectoral scopes linked to the applied methodologies	SS 1: Energy industries (renewable/non-renewable sources) SS 13: Waste handling and disposal
Project Verification Criteria: Mandatory requirements to be assessed	☒ UCR Standard ☒ Applicable Approved Methodology ☒ Applicable Legal requirements /rules of host country ☒ Eligibility of the Project Type ☒ Start date of the Project activity ☒ Meet applicability conditions in the applied methodology ☒ Credible Baseline ☒ Do No Harm Test

	☒ Emission Reduction calculations ☒ Monitoring Report ☒ No GHG Double Counting ☐ Others (please mention below)
Project Verification Criteria: Optional requirements to be assessed	☒ Environmental Safeguards Standard and do-no-harm criteria ☒ Social Safeguards Standard do-no-harm criteria
Project Verifier's Confirmation: The *UCR Project Verifier* has verified the UCR project activity and therefore confirms the following:	The UCR Project Verifier KBS Certification Services Ltd., certifies the following with respect to the UCR Project Activity Integrated Municipal Solid Waste Management Activities Programme, Nashik, India. ☒ The Project Owner has correctly described the Project Activity in the Project Concept Note (dated 07/11/2025) including the applicability of the approved methodology ACM0022: Consolidated Methodology: Alternative waste treatment processes, version 03.0 and AMS I.D: : Grid connected renewable electricity generation, version 18.0 meets the methodology applicability conditions and has achieved the estimated GHG emission reductions, complies with the monitoring methodology and has calculated emission reductions estimates correctly and conservatively.

	☒ The Project Activity is likely to generate GHG emission reductions amounting to the estimated (actual calculated) 1,099,221 tCO_2eq, as indicated in the PCN, which are additional to the reductions that are likely to occur in absence of the Project Activity and complies with all applicable UCR rules, including ISO 14064-2 and ISO 14064-3. ☒ The Project Activity is not likely to cause any net-harm to the environment and/or society ☒ The Project Activity complies with all the applicable UCR rules[1] and therefore recommends UCR Program to register the Project activity with above mentioned labels.
Project Verification Report, reference number and date of approval	GHG.25.VER.005_UCR Version 3.0 Date: 03/06/2026
Name of the authorised personnel of UCR Project Verifier and his/her signature with date	 Mr. Kaushal Goyal Managing Director

PROJECT VERIFICATION REPORT

Executive summary

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KBS Certification Services Ltd. has been contracted by "Nashik Waste Management Pvt. Ltd. (NWMPL)" to undertake the first verification and certification for the greenhouse gas (GHG) emission reductions registered under UCR, titled "Integrated Municipal Solid Waste Management Activities Programme, Nashik, India", UCR Ref. no. 492. This verification falls under the first monitoring period from 15/02/2017 to 31/12/2024 (both dates included), with the fixed crediting period between 15/02/2017 to 31/12/2024. The UCR projects must undergo independent third-party verification and certification of emission reductions as the basis for the issuance of Carbon Offset Units (COUs).

Verification Objectives and Scope:

The objectives of this verification exercise are, by reviewing objective evidence, to establish that:

- The project activity has been implemented and operated as per the approved project concept note /01/, and all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- The monitoring report /02/ and other supporting documents are complete;
- The data is recorded and stored as per the monitoring methodology and approved monitoring plan.
- To confirm that the monitoring system is implemented and fully functional to generate Carbon Offset Units (COUs) without any double counting.

The scope of the verification is the independent and objective review and ex-post determination of the monitored reductions in GHG emission by the project activity. The verification is based on the review of the monitoring report, supporting information and

a) The latest PCN /01/;
b) Monitoring report /02/ for the monitoring period under verification including COU calculations sheets and all supporting documents;
c) The applied monitoring methodology /05/;
d) Relevant decisions, clarifications, and guidance from UCR /04/;
e) All information and references relevant to the project activities resulting in emission reductions

KBS has assessed project activity against the requirements of the latest version of UCR verification standard, version 2.0 /04/, and employed a risk-based approach to verification, focusing on the identification of significant risks for project implementation and the generation of emission reductions.

Description of the Project:

The objective of the project activity is to register the Municipal Solid Waste Management Activities under the UCR activity. The project is implemented and operated by Nashik Waste Management Private Limited (NWMPL). The project aims to process and scientifically manage the entire municipal solid waste generated in Nashik city to reduce greenhouse gas emissions that would otherwise occur from open dumping and landfilling. The integrated waste management facility utilizes multiple technologies to convert waste into reusable and renewable resources, thereby promoting circular economy principles and mitigating climate change.

The project has been implemented and operated in accordance with the information in the approved PCN. The project is currently undergoing verification, and the monitoring period for the registered project activity is from 15/02/2017 to 31/12/2024 (both dates inclusive). The total verified emission reductions claimed under this monitoring period as verified are 1,099,221 tonnes of CO_2e.

Verification process:

The verification comprises a detailed review of the monitoring report covering the monitoring period from 15/02/2017 to 31/12/2024 (both dates inclusive) including monitoring parameters and monitoring plan, emission reduction calculation spread-sheet, monitoring methodology, and all related evidence provided by the project participant.

Methodology:

KBS follows a rule-based verification approach, wherein as a first step, a contract review is undertaken as per the latest version of the UCR Standard, version 7.0 /04/. A comprehensive desk review of the submitted project documentation, which is followed by an on-site verification visit by the members of the verification team in accordance with the latest version of the UCR Verification Standard, version 2.0 /04/. The project applies the approved CDM methodologies: ACM0022, version 03.0 and AMS-I. D, version 18.0 /05/. The verification protocol provides transparent means to record the observations and compliances by the verification team members and the nonconformities, if any. The verification protocol is an internal document and is available on request.

Conclusion:

Based on the verification assessment and subject to the successful closure of all findings, KBS confirms that the project activity has been implemented and operated in accordance with the approved PCN by NWMPL. All physical components of the project, including technology, project equipment, and monitoring and metering systems, are in place.
The monitoring systems, procedures, and emission reduction calculations are found to be in compliance with the requirements of the approved monitoring plan and the applicable monitoring methodology. Based on information reviewed and evaluated, we confirm that the implementation of the project has resulted in 1,099,221 tCO_2e of emission reductions during the monitoring period from 15/02/2017 to 31/12/2024, (both dates inclusive).

Project Verification team, technical reviewer and approver

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Project Verification team

No.	Role	Last name	First name	Affiliation (e.g. name of central or other office of UCR Project Verifier or outsourced entity)	Involvement in		
					Doc review	On-Site inspection	Interviews
1.	Team Leader and Technical Expert (TA 1.2)	Madan	Rishabh	Central Office	✓	✓	✓
2.	Technical Expert (TA 13.1)	Yadav	Ashish	Central Office	✓	✓	✓
3.	Verifier	Thomas	Alen Mariyam	Central Office	✓	✓	✓

Technical reviewer and approver of the Project Verification report

No.	Role	Type of resource	Last name	First name	**Affiliation** (e.g. name of central or other office of UCR Project Verifier or outsourced entity)
1.	Technical reviewer	IR	N. Urs	Praveen	Central Office
4.	Approver	IR	Goyal	Kaushal	Central Office

Means of Project Verification

Desk/document review

A desk review is undertaken, involving, but not limited to,

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, the quality of metering equipment, and the quality assurance and quality control procedures;
- An evaluation of data management and the quality assurance and quality control systems in the context of their influence on the generation and reporting of emission reductions.
- A complete list of documents evidence reviewed or referred in this report are included.

On-site inspection

Date of on-site inspection: 23/07/2025 to 25/07/2025

No.	Activity performed On-Site	Site location	Date
1)	The project verification team conducted interviews with the project owner to confirm the information and to resolve issues identified in the document review.	Nashik	23/07/2025 to 25/07/2025
2)	An assessment of the implementation and operation of the project activity as per the PCN and UCR requirements.		
3)	To validate that the project design, as documented is sound and reasonable, and meets the identified criteria UCR Standard Requirements and associated guidance		
4)	To assess conformance with the certification criteria as laid out in the UCR Standards;		
5)	To evaluate the conformance with the certification scope, including the GHG project and baseline scenarios; GHG sources, sinks, and reservoirs; and the physical infrastructure, activities, technologies and processes of the GHG project to the requirements of the GCC;		
6)	To evaluate the calculation of GHG emissions, including the correctness and transparency of formulae and factors used; assumptions related to estimating GHG emission reductions; and uncertainties; and		
7)	To determine whether the project could reasonably be expected to achieve the estimated GHG reduction/ removals.		
8)	A review of information flows for generating, aggregating and reporting of the ex-ante monitoring parameters.		
9)	Interviews with relevant personnel to confirm that the operational and data collection procedures can be implemented in accordance with the Monitoring Plan		
10)	A cross-check between information provided in the submitted documents and data from other sources		
11)	A review of calculations and assumptions made in determining the GHG data and estimated ERs, and		
12)	An identification of QA/QC procedures in place to prevent, or identify and correct, any errors or omissions in the reported monitoring parameters		

Interviews

No.	Interview			Date	Subject
	Last name	First name	Affiliation		
1.	Rege	Sameer	CEO, NWMPL	23/07/2025 to 25/07/2025	Project boundary, emission reduction calculations, monitoring plan (feasibility of monitoring arrangements described in PCN), QA/QC procedures, responsibility of implementation of monitoring plan, data recording & storage procedures, implementation plan.
2.	Mastaker	Dr. Priti	Consultant		
3.	Nagpurkar	Isha	Project Assistant		
4.	Nasker	Deboleena	Analyst		
5.	Kop	Viraj	DGM-Environment		
6.	Magdum	Sagar B	Lab Assistant, NWMPL		
7.	Ahire	Anil	Electrician, NWMPL		
8.	Kanande	Nandu	Fire & Safety / Security Incharge, NWMPL		
9.	Bagal	Sagar Bhima	Computer Operator, NWMPL		

Sampling approach

N/A

Clarification request (CLs), corrective action request (CARs) and forward action request (FARs) raised

Areas of Project Verification findings	No. of CL	No. of CAR	No. of FAR
Green House Gas (GHG)			
Identification and Eligibility of project type	-	-	-
General description of project activity	CL 01 CL 02 CL 03 CL 05	CAR 02	-
Application and selection of methodologies and standardized baselines	-	-	-
- Application of methodologies and standardized baselines	-	-	-
- Deviation from methodology and/or methodological tool	-	-	-
- Clarification on applicability of methodology, tool and/or standardized baseline	-	-	-
- Project boundary, sources and GHGs	CL 01	-	-
- Baseline scenario			-
- Estimation of emission reductions or net anthropogenic removals			-
- Monitoring Report	CL 03 CL 04 CL 05	CAR 03	-
Start date, crediting period and duration	-	CAR 01	-
Environmental impacts	-	-	-
Project Owner- Identification and communication	-	-	-
Others (please specify)	-	-	-
Total	**05**	**03**	**-**

Project Verification findings

Identification and eligibility of project type

Means of Project Verification	The project has been approved for verification under the UCR program with project reference number 492. (https://www.ucarbonregistry.io/Registry/Details?id=QjkmOLwGO3sR1G%2BIlHrD5Q%3D%3D). The project has taken reference with the approved CDM methodologies ACM0022: Alternative waste treatment processes, version 3.0 and AMS-I. D: Grid connected renewable electricity generation, version 18.0 /05/. The monitoring report complies with the approved PCN and the UCR Verification standard, version 2.0 /04/.
Findings	No findings were raised.
Conclusion	The verification team confirms that the project is in line with the UCR standard version 7.0 /04/, UCR verification standard version 2.0 /04/, and UCR program manual version 6.1 /04/.

General description of project activity

<table>
<tr><td>Means of Project Verification</td><td>
The objective of the project activity is to implement an Integrated Municipal Solid Waste Management (MSWM) system for Nashik City, Maharashtra, for the scientific processing, treatment, and disposal of municipal solid waste that would otherwise undergo uncontrolled anaerobic decomposition in open dumpsites. Nashik Waste Management Private Limited (NWMPL) is responsible for implementing and operating the project. The implementation of the project activity results in measurable GHG emission reductions compared to the baseline scenario. The project facility is located at Latitude 19.933564 and Longitude 73.739300.
The project has been implemented and operated in accordance with the information provided in the PCN /01/. The monitoring period undergoing verification spans from 15/02/2017 to 31/12/2024 (both dates inclusive). The total emission reductions claimed for this monitoring period amount to 1,099,221 tCO_2e.
The facility scientifically processes mixed municipal solid waste that would otherwise be landfilled and emit significant methane and carbon dioxide. The project converts waste into valuable resources through aerobic composting, RDF production, leachate biomethanation for biogas and electricity generation, plastic-to-fuel conversion, biomass briquetting, scientific landfilling, solar energy generation, and biomining of legacy waste. These interventions collectively reduce GHG emissions and support long-term sustainable waste management in Nashik.
The details of the project activities along with the start dates are as follows:
<table>
<tr><th>Sl. No</th><th>Title of Project Activities</th><th>Starting Date of Project Activity</th><th>Last Date of the First Crediting Period</th></tr>
<tr><td>0</td><td>Pre-Sorting</td><td>15/02/2017</td><td>31/12/2024</td></tr>
<tr><td>1</td><td>Composting</td><td>15/02/2017</td><td>31/12/2024</td></tr>
<tr><td>2</td><td>RDF Production</td><td>15/08/2017</td><td>31/12/2024</td></tr>
<tr><td>3</td><td>Leachate Treatment Plant (LTP)/ Biomethanation</td><td>15/02/2017</td><td>31/12/2024</td></tr>
</table>
</td></tr>
</table>

	<table><tr><td>4</td><td>Biomass Briquetting</td><td>15/04/2022</td><td>31/12/2024</td></tr><tr><td>5</td><td>Nashik Plastic to Fuel</td><td>19/03/2023</td><td>31/12/2024</td></tr><tr><td>6</td><td>Dead Animal Incinerator</td><td>15/02/2017</td><td>31/12/2024</td></tr><tr><td>7</td><td>Reject To Landfill</td><td>15/06/2018</td><td>31/12/2024</td></tr><tr><td>8</td><td>Solar Energy</td><td>13/09/2024</td><td>31/12/2024</td></tr><tr><td>9</td><td>Biomining and capping of existing Legacy Waste</td><td>15/02/2017</td><td>31/12/2024</td></tr></table> The facility receives waste collected by the Nashik Municipal Corporation and processes biodegradable fractions into compost through controlled aerobic treatment, while leachate is anaerobically treated to produce biogas for captive electricity and heat generation. Combustible fractions are converted into Refuse-Derived Fuel (RDF), plastic waste is processed into fuel through pyrolysis, and horticultural waste is converted into biomass briquettes as a renewable substitute for coal. The project also includes a controlled dead animal carcass incinerator operating on RDF and plastic-derived fuel, a scientific landfill for inert waste disposal, and systematic biomining of legacy waste accumulated since 2000 to recover materials and reclaim landfill space. Solar photovoltaic systems installed on-site supply renewable electricity through a net-metering arrangement, reducing reliance on fossil-fuel-based grid power. During the site visit and desk review, the verification team assessed the operational status of all major processing systems and the single line diagram. Records related to equipment calibration, maintenance, process control, waste inflows, energy generation, and fuel substitution were cross-checked against plant logs, metering records, weighbridge receipts, and invoices. Leakage sources, including transportation and auxiliary electricity use, were also evaluated. The verification team confirms that the project has been implemented in accordance with the approved PCN /01/ and that the monitoring procedures and parameters have been consistently and accurately applied throughout the monitoring period. The team further confirms that the project description is consistent with the registered design and that all physical features, equipment, and monitoring systems are in place and functioning as required.
Findings	CL 01, CL 02, CL 03, CL 05 and CAR 02 were raised and closed successfully. Refer to “Clarification request, corrective action request and forward action request” below for further details.
Conclusion	The verification team confirms that the project description contains all the relevant information required and is in line with the UCR standard version 7.0 /04/, UCR verification standard version 2.0 /04/, and UCR program manual version 6.1 /04/.

Application and selection of methodologies and standardized baselines

(.a.i) Application of methodology and standardized baselines

Means of Project Verification	The project activity applies to the approved CDM methodologies: ACM0022: Consolidated Methodology For Alternative Waste Treatment Processes, version 3.0 /05/ and AMS-I.D: Small-scale Methodology for Grid-connected Renewable Electricity Generation, version 18.0 // and for Project baseline emissions calculations for biomining of legacy waste, Chapter 3, Solid Waste Disposal, 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Microsoft Word -

	V5_Ch3_SWDS_Final_v2.doc is used. Supporting methodological tools were also used to calculate baseline and project emissions, and assess emissions from solid waste disposal, composting, and anaerobic digestion. The applicability of the applied methodologies are assessed below:

Applicability Condition under ACM0022, version 03	Verification team assessment
(a) Composting process under aerobic conditions;	The project activity treats biodegradable MSW through aerobic composting. The verification team checked this during desk review and on-site audit. The verification team confirms that the criteria has been met.
(b) Anaerobic digestion with biogas recovery and flaring and/or its use;	The project activity involves the treatment of liquid and residual solid waste through anaerobic digestion with biogas recovery, and does not include waste treatment processes other than anaerobic digestion, biogas recovery, and flaring/use. The same has been confirmed by verification team during desk review and on-site audit.
(c) Co-composting of wastewater in combination with solid waste;	The condition for co-composting of wastewater in combination with solid waste is not applicable.
(d) Anaerobic co-treatment of wastewater in combination with solid waste;	The condition for anaerobic co-treatment of wastewater in combination with solid waste is not applicable.
(e) Mechanical/thermal treatment process to produce refuse-derived fuel (RDF) or stabilized biomass (SB) that is produced within the project boundary and its use	The verification team confirms that RDF and SB are produced from waste originating within the project boundary and that, in the absence of the project, this waste would have remained in the SWDS and generated methane. The verification team confirms that this meets the applicability conditions of ACM0022, version 3.0.
(f) Gasification process to produce syngas and its use;	The condition for gasification process to produce syngas and its use is not applicable.
(g) Incineration of fresh waste for the generation of thermal/electric energy.	The condition for incineration of fresh waste for the generation of thermal/electric energy is not applicable.
(h) The following conditions apply to all project activities using this methodology:	-
(i) The project plant only treats	The verification team confirms

	fresh waste/wastewater for which emission reductions are claimed, except for cases involving composting, co-composting and anaerobic digestion;	that emission reductions are claimed solely for fresh waste, which is received daily and processed on the same day; hence, the applicability criteria are met, as verified during the on-site audit.
	(j) Neither the fresh waste nor the products from the project plant are stored on-site under anaerobic conditions;	The verification team confirms that measures are in place to ensure no on-site storage of fresh waste or project products under anaerobic conditions.
	(k) Any wastewater discharge resulting from the project activity is treated in accordance with applicable regulations;	The verification team confirms that the project claims emission reductions only for fresh waste, which is received and processed on the same day, and verifies that the applicability criteria are met based on the on-site audit.
	(l) The project activity does not reduce the amount of waste that would be recycled in the absence of the project activity. This shall be justified and documented in the clean development mechanism project design document (PCN);	The verification team confirms, based on the on-site audit, that recyclable waste from the fresh waste received at NWMPL is fully separated and sent for recycling, and that the applicability criteria are met.
	(m) When applicable regulations mandate any waste treatment process implemented under the project activity, the rate of compliance with such regulations for the treatment process is below 50 per cent;	The condition regarding compliance with applicable regulations, where the rate of compliance is below 50%, is not applicable.
	(n) Hazardous wastes/wastewater are not eligible under this methodology.	Based on the on-site audit, the verification team confirms that NWMPL treats only municipal solid waste and does not receive any hazardous waste; thus, the applicability criteria are met.
	The methodology is only applicable if the baseline scenario is:	-
	(a) The disposal of the fresh waste in a SWDS with or without a partial LFG capture system (M2 or M3)3;	Fresh waste coming to the site is processed daily.There is no LFG capture system present yet. There is no facility to capture LFG in the plant.
	(b) In the case of co-composting or co-treatment of wastewater in an anaerobic digester, the treatment of organic wastewater in either an existing or new anaerobic lagoon or sludge pit without methane recovery (W1 or W4);	The condition regarding treatment of organic wastewater in an anaerobic lagoon or sludge pit without methane recovery in the case of co-composting or co-treatment is not applicable.
	(c) In the case of electricity generation, the electricity is	The plant is connected through grid. No electricity is generated

generated in an existing/new captive fossil fuel fired power-only plant, captive cogeneration plant and/or the grid (P2, P4 or P6);	through the waste.
(d) In the case of heat generation, the heat is generated in an existing/new fossil fuel fired cogeneration plant, boiler or air heater (H2 or H4).4	RDF generated inside the project facility is used for heat generation in Dead Animal Carcass Incineration Process

Applicability Condition under AMS I.D., version 18.0	**Justification**
This methodology comprises of renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass for: Supplying electricity to the grid.	The project involves renewable energy generation through the installation of solar photovoltaic modules and supplies electricity to the grid, thereby replacing state grid electricity as verified during the on-site audit and desk review, Hence, the verification team confirms that this applicability criterion is met.
2. Illustration of methodology under ASM-I.D.	The verification team confirms that the project involves renewable energy generation through the installation of solar photovoltaic modules supplying electricity to the grid, is in line with the ASM-I.D. methodology, and that the applicability criterion is met.
3. This methodology is applicable to project activities that: Install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity, that is a Greenfield plant.	The project involves the installation of solar panels at sites where no renewable energy power plant was operational prior to the implementation of the project activity and qualifies as a Greenfield plant. Hence, the verification team confirms that the applicability criterion is met.
4. If the new unit has both renewable and non- renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The project involves only renewable solar energy, and the total capacity of the project is less than or equal to 15 MW, complying with the eligibility limit for a Type-I small-scale project activity. Hence, the verification team confirms that the applicability criterion is met.
5. Combined heat and power (co-generation) systems are not eligible under this category.	The verification team confirms that no cogeneration system is implemented in the component activity, which involves solar power. Accordingly, this eligibility requirement is not applicable.

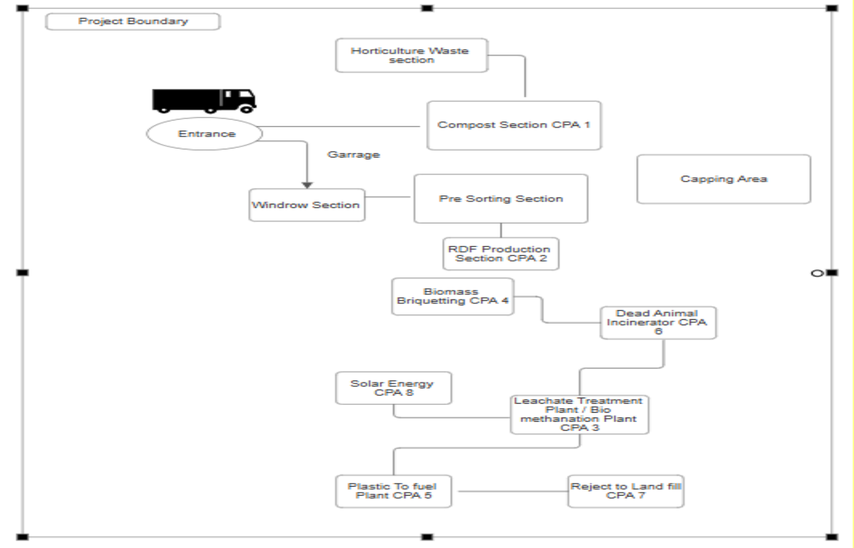
	<table><tr><td>6. In the case of project activities that involve the addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW, and should be physically distinct from the existing units.</td><td>Not applicable as the project activity involves solar power plants implementation.</td></tr><tr><td>7. In the case of project activities that involve the addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.</td><td>Not applicable as the project activity involves solar power plants implementation.</td></tr><tr><td>8. In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15 MW.</td><td>The verification team confirms that the project involves the installation of a new solar power plant with a capacity below 15 MW and does not constitute a retrofit or replacement unit. Hence, this applicability criterion is not applicable.</td></tr></table> The verification team has checked the applicability conditions with the applied methodology and tools and approved PCN and found to be correct.
Findings	No findings were raised.
Conclusion	"The verification team confirms that the applicability of the project is in line with the applied methodology, UCR Standard Version 7.0, UCR Verification Standard Version 2.0, UCR Program Manual Version 4.0, and the approved PCN.

(.a.ii) Clarification on applicability of methodology, tool and/or standardized baseline

Means of Project Verification	Not applicable.
Findings	Not applicable.
Conclusion	Not applicable.

(.a.iii) Project boundary, sources and GHGs

Means of Project Verification	The project boundary includes the physical, geographical site(s) of NWMPL:

	 The verification team has reviewed the project boundary in accordance with the approved PCN and confirms that it aligns with the methodology.
Findings	CL 01 was raised and closed successfully. Refer to “Clarification request, corrective action request and forward action request” below for further details.
Conclusion	The verification team confirms that the project boundary includes all necessary information and complies with both UCR requirements and the approved PCN.

(.a.iv) Baseline scenario

Means of Project Verification	**Baseline emissions:** Baseline emissions are determined as follows: $BE_y = \sum_t (BE_{CH4,t,y} + BE_{WW,t,y}) \times (1 - RATE_{compliance,t})$ Equation (1) Where, BE_y = Baseline emissions in year y (t CO_2e) $BE_{CH4,t,y}$ = Baseline emissions of methane from SWDS in year y (t CO_2e) $BE_{WW,t,y}$ = Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year y (t CO_2e) $RATE_{compliance,t}$= Discount factor to account for the rate of compliance of a regulatory requirement that mandates the use of alternative waste treatment process t. The verification team concludes that the PP has correctly applied the baseline methodologies and that the baseline scenario for the NWMPL project has been accurately established. In addition, several other activities contribute to baseline emissions. The total baseline emissions, hence, are given as follows: $BE = BE_y + BE_{RDF} + BE_{AD} + BE_{PtoF} + BE_{Briq} + BE_{solar} + BE_{Biomining}$ Equation (2) Where, BE = Total baseline emissions in year y (t CO_2e) BE_y = Baseline emissions in year y (t CO_2e) BE_{RDF} = Baseline emissions due to RDF use instead of coal in year y (t

CO_2e)
BE_{AD} = Baseline emissions avoided due to use of electricity produced from biogas in year y
(t CO_2e)
BE_{PtoF} = Baseline emissions avoided by using pyro oil produced in Plastic to fuel process in
year y. (t CO_2e)
$BE_{Biomining}$ = Baseline emissions avoided due to biomining (t CO_2e)

(a)Baseline emissions of methane from the SWDS ($BE_{CH4,t,y}$):

Baseline methane emissions from the SWDS are determined using —TOOL04: Emissions from solid waste disposal sites.

The emissions are calculated using the formula given below:

$$BE_{CH4,SWDS,y} = \emptyset_y \times (1 - f_y) \times GWP_{CH4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^{y} \sum_j (W_{j,x} \times DOC_j \times e^{-k(y-x)} \times (1 - e^{-k_j}))$$

Equation (3)

Where, for the yearly model

$BE_{CH4,SWDS,y}$ Baseline, project or leakage methane emissions occurring in year y
$PE_{CH4,SWDS,y}$ = generated from waste disposal at a SWDS during a time period ending
$LE_{CH4,SWDS,y}$ in year y (t CO_2e/yr).

X = Years in the time period in which waste is disposed at the SWDS,
extending from the first year in the time period (x = 1) to year y (x = y)
y = Year of the crediting period for which methane emissions are calculated
(y is a consecutive period of 12 months)
$DOC_{f,y}$ = Fraction of degradable organic carbon (DOC) that decomposes under the
specific conditions occurring in the SWDS for year y (weight fraction)
$W_{j,x}$ = Amount of solid waste type j disposed or prevented from disposal in the
SWDS in the year x (t).

$\emptyset$ = Model correction factor to account for model uncertainties for year y.
f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere
year y.
GWP_{CH4} = Global Warming Potential of methane
OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the material soil or other covering the waste)
J = Type of residual waste or types of waste in the MSW

F = Fraction of methane in the SWDS gas (volume fraction).
MCF_y = Methane correction factor for year y.
DOC_j = Fraction of degradable organic carbon in the waste type j (
K = Decay rate for the waste type j (1 / yr).

Given equation consists of two parts (part A and part B) as represented below:

$$BE_{CH4,SWDS,y} = \underbrace{\emptyset y \times (1 - fy) \times GWP_{CH4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y}_{A} \times \underbrace{\sum_{x=1}^{y} \sum_j (W_{j,x} \times DOC_j \times e^{-k_j(y-x)} \times (1 - e^{-k_j}))}_{B} \tag{3.1}$$

The values of the parameters in this equation are mentioned in the table below:

Parameter	SI Unit	Description	Value	
φy	ton/yr or m3/yr	Model correction factor to account for model uncertainties for year y	Default value	1
fy	tC/mass	Weighted average mass fraction of carbon in fuel type i in year y		0
$GWPCH4$	Mass unit/volume unit	Weighted average density of fuel type i in year y	Default value	21
OX	GJ/m³, GJ/ton	Weighted average net calorific value of fuel type i in year y	Default value	0.1
F	Volume fraction	Fraction of methane in the SWDS gas	Default value	0.5
$DOCf$,y	Weight fraction	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)	Default value	0.5
$MCFy$		Methane correction factor for year y	Default value	1
Wj,x	tonnes	Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x	Calculate using the waste characterisation	
$DOCj$	weight fraction	Fraction of degradable organic carbon in the waste type j	Default value	
k	1 / yr	Decay rate for the waste type j	Default value	
j		Type of residual waste or types of waste in the MSW		
x		Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x		

		= y)		
y		Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)		

Part A of the equation is calculated using default values of the parameters and part B is calculated using the plant data measured by following the monitoring procedures.

Part A:

Determination of the methane correction factor (MCFy)

The MCF is selected as a default value ($MCF_y = MCF_{default}$) provided in the section —Data and parameters not monitored of the PCN.

Determining the model correction factor (φ_y)

The model correction factor (φ_y) depends on the uncertainty of the parameters used in the FOD model. For Baseline emissions the value applicable to application B as per the location of the project is φ_y = 0.85.

All the parameters of the part A are default values as given below:

φy	1
fy	0
1-fy	1
GW_{CH4}	21
OX	0.1
1-OX	0.9
F	0.5
MCF_y	1
$DOC_{f,default}$	0.5
A	5.4

Part B:

Determining the amounts of waste types j disposed in the SWDS ($W_{j,x}$ or $W_{j,i}$)

$W_{j,x}$ is the weight of organic waste type j disposed in SWDS in a year.

Where different waste types j, are disposed or prevented from disposal in the SWDS (for example, in the case of MSW), it is necessary to determine the amount of different waste types ($W_{j,x}$ or $W_{j,i}$).
The amount of different waste types is determined through sampling and the mean is calculated from the samples using equation (5) to determine the value of $W_{j,x}$ for the yearly model as follows:

$$W_{j,x,x} = W_x + p_j \qquad \text{Equation (4)}$$

Where:

= Amount of solid waste type j disposed or prevented from disposal in the SWDS

W_x = Total amount of solid waste disposed or prevented from disposal in the SWDS

$p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction).

J = Type of residual waste or types of waste in the MSW

X = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)

The fraction of the waste type j in the waste for the year x are calculated according to equation (4) as follows:

$$p_{j,x} = \frac{\sum_{n=1}^{z_x} p_{n,j,x}}{z_x} \qquad \text{Equation (5)}$$

$p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction).

z_x = Number of samples collected during the year x.

$p_{n,j,x}$ = Fraction of the waste type j in the sample n collected during the year x (weight Fraction)

Determining the fraction of DOC that decomposes in the SWDS ($DOC_{f,y}$):

Application A

$DOC_{f,y}$ is given as a default value ($DOC_{f,y}$ = $DOC_{f,default}$) provided in the section —Data and parameters not monitored of the PCN.

Plant data provided the characterisation of incoming waste, with four samples collected annually. Each sample consists of approximately 100 kg of mixed waste, which is sorted into six categories: Wood and wood products; Pulp, paper, and cardboard (excluding sludge); Food, food waste, beverages, and tobacco (excluding sludge); Textiles, Garden, yard, and park waste; and Glass, plastic, metal; and other inert waste. The degradable organic carbon fraction (DOC_j) in each waste type is a critical parameter used to estimate carbon emissions. Each waste type has a default DOC_j value, as shown in the accompanying table.

Default values for *DOC_j*

Waste type *j*	***DOC_j* (% wet waste)**
Wood and wood products	43
Pulp, paper and cardboard (other than sludge)	40
Food, food waste, beverages and tobacco (other than sludge)	15
Textiles	24
Garden, yard and park waste	20
Glass, plastic, metal, other inert waste	0

Source: IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)

Another critical parameter in the formula for estimation of carbon emissions is the decay rate for the waste type j. Following table gives the default value of this parameter for different waste types.

Default values for the decay rate (k_j)

Waste type j		Boreal and Temperate (MAT≤20°C)		Tropical (MAT>20°C)	
		Dry (MAP/PET <1)	Wet (MAP/PET >1)	Dry (MAP< 1000mm)	Wet (MAP > 1000 mm)
Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
	Wood, wood products and straw	0.02	0.03	0.025	0.035
Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06	0.185	0.085	0.40

Source: IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)

Sample calculation for part B:

The amount of waste of a particular type is calculated based on the data gathered for quantity of incoming waste and waste characterisation samples. The waste characterisation is done by quadrant method. Total Weight of MSW is constantly measured digitally and recorded both manually and on digital system.

year	x	y	Waste type	Wj,x	DOCj	kj	y-x	e(-kj(y-x))	exp(-kj)	1-exp(-kj)	product	B	CO2 e t/yr
2017	1	1	Food, food waste, beverages and to	66541	0.15	0.4	0	1	0.67032	0.32968	3290.57	3560.54	3,560.54
			Garden, yard and park waste	5012	0.2	0.17	0	1	0.843665	0.156335	156.71		
			Pulp, paper and cardboard (other t	4189	0.4	0.07	0	1	0.932394	0.067606	113.27		
			Textiles	7366	0.24	0	0	1	1	0	0.00		
			Wood and wood products		0.43	0.035	0	1	0.965605	0.034395	0.00		
			Glass, plastic, metal, other inert wa	59290	0	0	0	1	1	0	0.00		
2018	1	2	Food, food waste, beverages and to	102587	0.15	0.4	1	0.67032	0.67032	0.32968	3400.61	4,772.90	11,344
			Garden, yard and park waste	8806	0.2	0.17	1	0.8436648	0.843665	0.156335	232.29		
			Pulp, paper and cardboard (other t	8139	0.4	0.07	1	0.9323938	0.932394	0.067606	205.23		
			Textiles	61789	0.24	0.07	1	0.9323938	0.932394	0.067606	934.77		
			Wood and wood products	0	0.43	0.035	1	0.9656054	0.965605	0.034395	0.00		
			Glass, plastic, metal, other inert wa	15527	0	0	1	1	1	0	0.00		
2018	2	2	Food, food waste, beverages and to	102587	0.15	0.4	0	1	0.67032	0.32968	5073.12	6,571.11	
			Garden, yard and park waste	8806	0.2	0.17	0	1	0.843665	0.156335	275.33		
			Pulp, paper and cardboard (other t	8139	0.4	0.07	0	1	0.932394	0.067606	220.11		
			Textiles	61789	0.24	0.07	0	1	0.932394	0.067606	1002.55		
			Wood and wood products	0	0.43	0.035	0	1	0.965605	0.034395	0.00		
			Glass, plastic, metal, other inert wa	15527	0	0	0	1	1	0	0.00		

As given above, the calculations are further carried out for a crediting period of 7 years.

The total baseline emissions of methane from the SWDS ($BE_{CH4,t,y}$) for 7 years crediting period are accounted to be 951915 t CO_2 e.

(b)Baseline emission from organic wastewater ($BE_{WW,t,y}$):

Baseline emissions are determined using equation (6). Out of the values of amount of methane emissions from wastewater and baseline methane emissions determined using the Methane Conversion Factor, whichever is minimum, is chosen as the Baseline methane emissions from anaerobic treatment of the wastewater in open.

$$BE_{WW,t,y} = \min\{Q_{CH4}; BE_{CH4,MCF,y}\}$$

Equation (6)

Where:

$BE_{\overline{W}W,t,y}$ Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year y (t CO_2e).

$Q_{CH4,MCF,y}$ = Amount of methane produced from wastewater in year y after the implementation of the project activity (t CO_2e).

$BE_{CH4,MCF,y}$ = Baseline methane emissions determined using the Methane Conversion Factor (tCO_2e)

As per equation 5, the baseline emissions are the total amount of methane produced from wastewater in year y after the implementation of the project activity for 7 years which is accounted to be 225 t CO_2 e.

Baseline methane emissions determined using the methane conversion factor ($BE_{CH4,\ MCF,y}$):

$$BE_{CH4,MCF,y} = GWP_{CH4} \times MCF_{BL,y} \times B_0 \times COD_{BL,y}$$ Equation (7)

Where:

$BE_{CH4,MCF,y}$ = Baseline methane emissions determined using the Methane Conversion Factor (tCO_2e)

GWP_{CH4} = Global Warming Potential of methane valid for the commitment period (t CO_2e/t $_{CH4}$)

$MCF_{BL,y}$ = Average baseline methane conversion factor (fraction) in year y, representing the fraction of ($COD_{BL,y}$ x Bo) that would be degraded to CH_4 in the absence of the project activity.

B_0 = Maximum methane producing capacity, expressing the maximum amount of CH_4 that can be produced from a given quantity of chemical oxygen demand (t CH_4/tCOD)

$COD_{BL,y}$ = Quantity of chemical oxygen demand that would be treated in anaerobic lagoons or sludge pits in the absence of the project activity in year y (t COD).

The total baseline emissions from organic wastewater ($BE_{WW,t,y}$) using methane correction factor for 7 years crediting period are accounted to be 0.03205 t CO_2 e.
The minimum value is of the methane emissions determined using the methane conversion factor ($BE_{CH4,MCF,y}$)
Hence the Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons in the absence of the project activity are 0.03205 t CO_2

$$BE_y = \sum_t (BE_{CH4,t,y} + BE_{WW,t,y}) \times (1 - RATE_{compliance,t})$$

Total Baseline emissions as per equation (1) are 759819 t CO_2 e for the crediting period of 7 years.

Baseline, project and leakage emissions contributed by some other processes are mentioned further.

(c)Anaerobic digestion/Leachate Biomethanation:

Leachate generated from various sections of the plant, including pre-sorting, sorting, and RDF processing is directed to the leachate biomethanation unit, where it is used for biogas production. The resulting biogas is then converted into electricity using a biogas generator (BG-set). This conversion is carried out in a batch process. The biogas is first stored in a balloon, and the BG-set is operated once the balloon is fully filled. The electricity produced is utilized for heating purposes in the incinerator.

The baseline emissions for Leachate bio-methanation have been calculated using tool 7 as follows:

$BE_{AD} = BE_{EC} = \sum EC_{BE,} \times EF_{EF}$ Equation (8)

In the absence of this project activity, coal-based electricity would have been used instead. The baseline emissions avoided through the use of this renewable electricity amount to 254 tCO_2e.

(d)RDF manufacturing.

The dry fraction of the daily incoming municipal solid waste (MSW) is utilized for the production of Refuse-Derived Fuel (RDF). The final product, RDF has a high calorific value, making it an effective alternative fuel for use in boilers. The RDF produced at this facility is partially used internally and partially supplied to various vendors who use it for heating purposes. By replacing coal, a high carbon-emitting energy source, with RDF, significant emission reductions are achieved.

The baseline emissions for RDF manufacturing have been calculated using tool 3 as follows:

$BE_{RDF} = \sum_i FC_{i,j,y} \times COEF_{i,y}$ Equation (9)

Where,

$FC_{i,j,y}$ = Is the quantity of fuel type i combusted in process j during the

	year y (mass or volume unit/yr) $CO_{EFi,y}$ = Is the CO_2 emission coefficient of fuel type i in year y (tCO2/mass or volume unit) In order to calculate $CO_{EFi,y,}$ the following formula is use: $COEF_{i,y} = NCV_{i,y} + EF_{CO2,i,y}$ Equation (10) Where, NCV_{RDF} = Is the weighted average net calorific value of the RDF in year y $EF_{CO2,yRDF}$ = Is the weighted average CO_2 emission factor of RDF in year y (tCO_2/GJ). Over a period of seven years, a total of 324,548 tonnes of RDF has been produced, resulting in the avoidance of approximately 590,029 tonnes of CO_2 emissions. These represent the baseline emissions savings. **(e)Incineration:** Dead animals are treated through incineration at the facility. In the absence of this activity, the carcasses would have been left to decompose naturally, leading to significant greenhouse gas emissions. According to research, a carcass weighing 1,000 kg emits approximately 858 kg of CO_2 through natural decomposition. Over a period of seven years, the facility has treated a total of 2,733,150 kg of dead animals, resulting in avoided emissions of approximately 2,345 tCO_2. These avoided emissions contribute to the baseline emission reductions associated with the project. **(f)Plastic to fuel:** Plastic to fuel is a pyrolysis process decomposing heavy plastics to produce pyro oil and pyro gas. The pyro gas produced here in this plant, is stored in a balloon and reused as a fuel for the pyrolysis chamber. The pyro oil produced here is also used in incineration. This pyro oil is replacing fuel in incineration. Considering the oil is replacing diesel, this process is avoiding 24 t CO_2. **(g)Solar:** The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid." The solar power plant supplements the facility's electricity needs alongside the existing grid supply. During the daytime, solar panels generate electricity, while at night, the facility draws power from the grid. To accurately monitor the flow of electricity, a bidirectional meter has been installed outside the project boundary to record both the import from and export to the grid. A company named Surya Logic has implemented the plant's control system, which tracks all data related to

	electricity generation, supply, and consumption through a dedicated mobile application. $BE_{EC,} = \sum EG_{PJ,} \times EF_{EF}$ Equation (11) $BE_{EC,}$ = Project emissions due to electricity consumption in year y (tCO_2e) $EG_{PJ,}$ = Net electricity supplied to the grid facility by the project activity $EF_{EF,y}$ = Emission factor for electricity generation in year y ($t\ CO_2/MWh$) The value of CO_2 emission for coal power grid plants is obtained by referring the UCR document "UCR COU Standard Update: 2024 Vintage UCR Indian Grid Emission Factor Announced The solar plant became operational on 15th September 2024. Within just three and a half months, the 0.39 MWH installation has successfully avoided approximately 57 tonnes of CO_2 emissions. **(h)Briquetting:** In addition to daily municipal solid waste (MSW), the plant also processes horticulture and garden waste. The weight data for this waste stream is documented in Annexure 4. Garden waste, being organic in nature, can emit significant amounts of greenhouse gases (GHGs) if left untreated or left to decompose in open fields. At the facility, this biomass is either converted into briquettes or, in some cases, dried and used directly as fuel for industrial combustion. Over a period of seven years, the treatment of 120,889 tonnes of horticulture waste has prevented an estimated 47893 tonnes of CO_2 emissions. These avoided emissions represent the baseline for this process. **(i)Baseline emissions avoided due to biomining (tCO_2e)** **Biomining** is a mechanical–biological process used for the remediation of legacy municipal solid waste dumpsites. It involves excavation, segregation, and processing of old waste to recover recyclables, inert materials, and soil-like fractions, while stabilizing or disposing of residuals in an environmentally sound manner. By removing the remaining degradable organic matter from anaerobic conditions, biomining halts future methane generation that would have occurred if the waste had remained in place. In this project, baseline methane emissions from the dumpsite were estimated using the IPCC 2006 First-Order Decay (FOD) model, and these are quantified as avoided emissions. Baseline emissions avoided due to biomining are calculated using the following steps: **Step1:** $DOC = \sum_i f_i \times DOC_i$ Equation (12) Where, DOC = fraction of degradable organic carbon in bulk waste, Gg C/Gg waste DOC_i = fraction of degradable organic carbon in waste type i e.g., the

	default value for paper is 0.4 (wet weight basis) W_i = fraction of waste type i by waste category e.g., the default value for paper in MSW in Eastern Asia is 0.188 (wet weight basis) **Step 2:** $DDOC_m = W \times DOC \times DOC_f \times MCF$ Equation (13) Where, $DDOC_m$ = mass of decomposable DOC deposited, t W = mass of waste deposited, t DOC = degradable organic carbon in the year of deposition, fraction, Gg C/Gg waste DOCf = fraction of DOC that can decompose (fraction) CH_4 correction factor for aerobic decomposition in the year of deposition (fraction) MCF = CH_4 correction factor for aerobic decomposition in the year of deposition (fraction) **Step 3:** $CH_4 Generated = DDOC_m \times F \times \frac{16}{12}$ Equation (14) Where, CH_4 generated = amount of CH_4 generated from decomposable material $DDOC_m$ = mass of decomposable DOC deposited, t F = fraction of CH_4, by volume, in generated landfill gas (fraction) 16/12 = molecular weight ratio CH_4/C (ratio) **Step 4:** $CH_4\ emissiosns = [\sum_x CH_4 Generated - R_T] \times (1 - OX_T)$ Equation (15) Where, CH_4 Emissions = CH_4 emitted in year T, t T = inventory year x = waste category or type/material R_T = recovered CH_4 in year T, Gg OX_T = oxidation factor in year T, (fraction), **Step 5:** $BE_{Bioimining} = CH_4\ emissions \times GWP$ Equation (16) Where, $BE_{Bioimining}$ = Baseline emissions avoided due to biomining $CH_4\ emissions$ = CH_4 emitted in year T, t GWP = Global warming potential of CH_4 Total waste mass present in the intervention area (W) = 80,244.5 t (based on contour mapping) f_i is the measured waste composition (mass fraction of food, paper,

	garden, wood, textiles) for the year 2017. F - fraction of CH_4, by volume, in generated landfill gas (fraction) is chosen as default value. R_T - recovered CH_4 in year T, Gg-The landfill gas is not recovered at the present facility MCF - CH_4 correction factor for aerobic decomposition in the year of deposition (fraction) is default value from IPCC OX_T - oxidation factor in year T, (fraction), is default value from IPCC GWP - Global warming potential of CH_4 (GWP for methane to be conservative as per UCR guidelines at 21 when applied for baseline calculations Total baseline emissions avoided due to biomining are 167760 t CO_2 e. For the baseline scenario, the PP has correctly identified the parameters and provided the necessary monitoring data, waste composition details, biogas and leachate generation data, electricity consumption records, and other supporting documentation required under the applied methodologies, including ACM0022, version 3.0, AMS-I.D, version 18.0, and associated tools. These parameters reflect the baseline conditions of untreated municipal solid waste, grid-based electricity consumption, and fuel use for waste handling in the absence of the project activity. The verification team cross-checked the supporting documents and confirms that all baseline values have been appropriately incorporated into the ER calculation sheet.
Findings	No findings were raised.
Conclusion	The verification team confirms that the baseline scenario has been correctly identified and is consistent with the approved methodology, UCR requirements, and the approved PCN.

(.a.v) Estimation of emission reductions or net anthropogenic removal

Means of Project Verification	In accordance with the applied methodology, the project owner has calculated emission reductions in the following manner: $ER_y = BE_y – PE_y – LE_y$ **Baseline emission calculations** Baseline emissions, estimated as per the formulas above, represent the total emissions that would have occurred in the absence of the project activity and primarily arise from uncontrolled anaerobic decomposition of municipal solid waste in open dumpsites, fossil-fuel-based grid electricity consumption, and fuel use in on-site transport. As untreated MSW releases significant quantities of methane, carbon dioxide, and nitrous oxide, baseline sources include emissions from SWDS, leachate biomethanation, RDF, incineration, plastic-to-fuel operations, solar displacement, briquetting, and biomining. These components collectively define the total baseline emissions against which the project's emission reductions are quantified. **Project emission calculations** Project emissions shall be calculated using the following equation: $PE_y = PE_{elec,y} + PE_{fuel,y} + PE_{Comp,y} + PE_{AD,y} + PE_{INC,y}$ Equation (17)

	Where, PE_y = Project emissions in year y (tCO_2e) $PE_{elec,}$ = Project emissions due to electricity consumption in year y (tCO_2e) $PE_{fuel,}$ = Project emissions due to fossil fuel consumption in year y (tCO_2e) $PE_{comp,}$ = Project emissions associated with composting in year y ($t\ CO_2e/yr$) $PE_{AD,}$ = Project emissions associated with the anaerobic digester in year y ($t\ CO_2e$) **a) Project emissions due to electricity consumption in year y ($PE_{elec,}$ tCO_2e)** The emissions include electricity consumption $PE_{elec,y}$ in the plant including leachate biomethanation, RDF manufacturing, composting, plastic to fuel, incineration, etc, calculated as per the "Tool to calculate baseline, project and/or leakage emissions from electricity consumption". Here the electricity is drawn from the national grid. For the calculation of $PE_{elec,}$ $PE_{EC,} = \sum EC_{PJ,} \times EF_{EF}$ Equation (18) $PE_{EC,}$ = Project emissions due to electricity consumption in year y (tCO_2e) $EC_{PJ,}$ = Quantity of electricity consumed by the project electricity consumption in year y (MWh/yr) $EF_{EF,y}$ = Emission factor for electricity generation in year y ($t\ CO_2/MWh$) Total PE from electricity consumption for monitoring period is = 3205 tCO_2e **(b)Project emissions due to fossil fuel consumption in year y (tCO_2e)** The emissions include fossil fuel consumption $PE_{fuel,y}$ associated with all the activities in the plant within the project boundary and diesel generator set which is used as a backup. Project emissions due to the fuel combustion are calculated as per the "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion". The project activity operates approx. 310 trucks for waste transport within the boundary and consists of two DG sets of capacity 400 kVA. Project emission due to fuel consumption is calculated as followed $PEFC_{j,y} = \sum_i FC_{i,j,y} \times COEF_{i,y}$ Equation (19) Where, PE_{FC}, = Project emissions due to fuel consumption in year y (tCO_2e) FC_i, = Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr) $COEF_{i,y}$ = Is the CO2 emission coefficient of fuel type i in year y (tCO_2/mass or volume unit) The quantity of diesel combusted in DG set and transportation during the year y is obtained from the purchased fuel invoices/bills, where the

	purchased fuel can be identified specifically for the CDM project. The CO_2 emission coefficient, is calculated based on net calorific value and CO_2 emission factor of the fuel type i, as follows: $COEF_{i,y} = NCV_{i,y} + EF_{CO2,i,y}$ Equation (20) Where: $COEF_{i,y}$ = Is the CO_2 emission coefficient of fuel type i in year y (tCO_2/mass or volume) $NCV_{i,y}$ = Is the weighted average net calorific value of the fuel type i in year y $EF_{CO2,j,y}$ = Is the weighted average CO_2 emission factor of fuel type i in year y (tCO_2/GJ) i = Are the fuel types combusted in process j during the year y Total PE from fossil fuel consumption = 519 tCO_2e **(c)Project emissions associated with composting in year y ($PE_{COMP,y}$ tCO_2e/yr)** MSW that is received every day is sorted into organic and inorganic waste. The organic waste is pre-sorted and sent to the composting unit. The UNFCCC ACM0022 methodology provides a standardized framework for calculating emissions from composting activities. Specifically, Tool 13 ("Project and leakage emissions from composting") outlines the procedures for determining both project and leakage emissions from composting and co-composting processes. In this case, only composting is conducted at the site; co-composting is not applicable. Composting results into project and leakage emissions. Following parameters are obtained using this methodology. The project emissions will be calculated using the below equation: $PE_{COMP,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH4,y} + PE_{N2O,y} + PE_{RO,y}$ Equation (21) Where, $PE_{COMP,y}$ = Baseline emissions of methane from the SWDS in year y (t CO_2e) $PE_{EC,y}$ = Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year y (t CO_2e). $PE_{FC,y\ y}$ = Project emissions from fossil fuel consumption associated with composting in year y (t CO_2/yr). $PE_{CH4,y}$ = Project emissions of methane from the composting process in year y (tCO_2e/yr) $PE_{N2O,y}$ = Project emissions of nitrous oxide from the composting process in year y (t CO_2e/yr) $PE_{RO,y}$ = Project emissions of methane from run-off wastewater associated with co-composting in year y (t) From the equation 21, $PE_{EC,y}$ and $PE_{FC,y}$ are excluded as they are already calculated earlier using equation 18,19 and 20. **Determination of project emissions of methane ($PE_{CH4,y}$)**

	For the estimation of $PE_{CH4,y}$ following equation is used: $PE_{CH4} = Q_y \times EF_{CH4,y} \times GWP_{CH4}$ Equation (22) Where: PE_{CH4} = Project emissions of methane from the composting process in year y (t CO_2e/yr). Q_y = Quantity of waste composted in year y (t/yr) EF_{CH4} = Emission factor of methane per tonne of waste composted valid for year y (t CH4 / t) GWP_{CH4} = Global Warming Potential of CH_4 (t CO_2e / t CH_4) In the above equation, the following parameters have default values: <table><tr><th>Parameter</th><th>value</th></tr><tr><td>$EF_{CH4,y}$</td><td>0.002</td></tr><tr><td>GWP_{CH4}</td><td>28</td></tr></table> Quantity of waste composted in year y (wet basis) is measured using a weighbridge. **Determination of project emissions of nitrous oxide (PEN_2O,y)** For estimating PEN_2O,y, following equation is used: $PE_{N2O,y} = Q_y \times EF_{N2O,y} \times GWP_{N2O}$ Equation (23) $PE_{N2O,y}$ = Project emissions of methane from the composting process in year y (t CO_2e/yr). Q_y = Quantity of waste composted in year y (t/yr) $EF_{N2O,y}$ = Emission factor of methane per tonne of waste composted valid for year y (t CH_4 / t) $GWP_{N2O,}$ = Global Warming Potential of CH_4 (t CO_2e / t CH_4) Global Warming Potential of N_2O value is obtained from the latest version of —Standard for application of the global warming potentials to clean development mechanism project activities and programmes of activities for the second commitment period of the Kyoto protocol‖. <table><tr><th>Parameter</th><th>value</th></tr><tr><td>$EF_{N2O,y}$</td><td>0.0002</td></tr><tr><td>GWP_{N2O}</td><td>265</td></tr></table> Calculations for the $PE_{RO,y}$: Project emissions of methane from run-off wastewater ($PE_{RO,y}$) are calculated only for the case of co-composting. In this case there is no co-composting. Project emissions amount associated with the composting for the period of 7 years =112375 tCO_2e. **(d) Project emissions associated with the anaerobic digester in year y (t CO_2e) / Leachate biomethanation (LB):** The leachate biomethanation process involves treating landfill leachate using anaerobic digestion to produce methane (CH_4). Methane is then either captured and utilized for energy generation (electricity production

or heating) or flared.
In the current case, methane is partially used for electricity generation and heating purposes and partially flared to prevent its release into the atmosphere.
We calculate the emissions from this activity, using flowing steps:

Parameter	SI Unit	Description
$PE_{AD,y}$	t CO_2e	Project emissions associated with the anaerobic digester in year y (t CO_2e)
$LE_{AD,y}$	t CO_2e	Leakage emissions associated with the anaerobic digester in year y (t CO_2e)

Project emissions calculation:

The project emissions will be calculated using the equation below:

$$PE_{AD,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH4,y} + PE_{flare,y} \qquad \text{Equation (24)}$$

Where:
$PE_{AD,y}$ = Project emissions associated with the anaerobic digester in year y (t CO_2e)
$PE_{EC,y}$ = Project emissions from electricity consumption associated with the anaerobic digester in year y (t CO_2e).
$PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO_2e).
$PE_{CH4,y}$ = Project emissions of methane from the anaerobic digester in year y (t CO_2e)
$PE_{flare,y}$ = Project emissions from flaring of biogas in year y (t CO_2e).

From the equation (19), PE_{EC},y and PE_{FC},y are excluded as they are calculated separately.

For $PE_{CH4,y}$, following equation is used:

$$PE_{CH4} = Q_{CH4} \times EF_{CH4,default} \times GWP_{CH4} \qquad \text{Equation (25)}$$

$Q_{CH4,y}$ = Amount of biogas collected at the digester outlet in year y .
$EF_{CH4,default}$ = Default emission factor for the fraction of CH_4 produced that leaks from the anaerobic digester which is 0.002 .
Global warming potential ($GWPCH_4$) is 28.
The gas flow is measured based on the capacity of the pump delivering the gas.

$PE_{flare,y}$ is calculated using Tool 8. Following equation provides emissions due to flaring based on mass flow measurement of gas in the minute.

However, at the plant site, annual volumetric gas flow measurement is given for biogas in m^3.

$$PE_{flare,y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

Equation (26)

Where:
$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO_2e)
GWP_{CH4} = Global warming potential of methane valid for the commitment period (t CO_2e/t)
$F_{CH4,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

	$\eta_{flare,m}$ = Flare efficiency in the minute m Mass flow of methane is obtained using tool 8. The equation provides mass flow of methane in the residual gas: $F_{i,t} = V_{t,db} \times v_{i,t,db} \times \rho_{i,t}$ Equation (27) Where: $F_{i,t}$ = Mass flow of greenhouse gas i in the gaseous stream in time interval t (kg gas/h) $V_{t,db}$ = Volumetric flow of the gaseous stream in time interval t on a dry basis (m^3 dry gas/h) $V_{i,t,db}$ = Volumetric fraction of greenhouse gas i in the gaseous stream in a time interval t on a dry basis (m^3 gas i/m^3 dry gas) $\rho_{i,t}$ = Density of greenhouse gas i in the gaseous stream in time interval t (kg gas i/m^3 gas i) Volumetric fraction of methane in biogas is 0.6. Density of methane in biogas is 0.716 kg/m3. The mass flow of gas is measure based on the capacity of the pump delivering the gas. Calculations are explained on page _final step of the tool 14. $PE_{flare,y} = BG_{flare,y} \times w_{CH4} \times GWP_{CH4} \times (1 - CEF)$ Equation (28) For simplification, the emissions from flaring are calculated based on annual data instead of minutes. Following equation is use. Where: $PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO_2e $BG_{flare,y}$ = Amount of biogas flared (kg) W_{CH4} = Fraction of methane in gas generated in digester (0.6) GWP_{CH4} = Global warming potential of methane (28) CEF= Combustion efficiency for open flare (0) Project emissions due to leachate bio-methanation for 7 years are 170 tCO_2 **Incineration:** In the incineration process, dead animal carcass is fed to the incinerator. Pyro oil, RDF and shredded biomass are used as a fuel to burn dead animal waste. Biogas from leachate biomethanation plant is used as a fuel for electricity generation. This electricity is used in incineration to heat up the feed. The gases of this completed combustion are passed through venturi scrubber where particulate matters, SO_2 and HCL are stripped off with the help of alkaline water. Project emissions from this process of incineration are obtained using the following equation: $PE_{INC,y} = PE_{COM,y} + PE_{EC,y} + PE_{FC,y} + PE_{WW,y}$ Equation (29) Where, $PE_{INC,y}$ = Project emissions from incineration in year y (tCO_2e). $PE_{COM,y}$ = Project emissions from combustion within the project boundary of fossil waste associated with incineration in year y (tCO_2). $PE_{EC,y}$ = Project emissions from electricity consumption associated with incineration in year y (tCO_2e). $PE_{FC,y}$ = Project emissions from fuel combustion associated with incineration in year y (tCH_4) $PE_{WW,y}$ = Project emissions from the wastewater treatment associated with incineration in year y. $PE_{COM,y} = PE_{COM,CO2,y} + PE_{COM,CH4,N2O,y}$ Equation (30)

	Where: $PE_{COM,y}$ = Project emissions from combustion within the project boundary associated with combustor c in year y (t CO_2e) $PE_{COM,CO2,y}$ = Project emissions of CO_2 from combustion within the project boundary associated with combustor c in year y (t CO_2) $PE_{COM,CH4,N2O,y}$ = Project emissions of CH_4 and N_2O from combustion within the project boundary associated with combustor c in year y (t CO_2). c = Combustor used in the project activity: gasifier or syngas burner, incinerator RDF/SB combustor **Project emissions of CO_2 from combustion within the project boundary (PECOM_CO_2,c,y):** $PE_{COM,CO2,c,y} = \frac{44}{12} \times FF_{COM,y} \times Q_{waste,c,y} \times FFC_{waste,c,y}$ Equation (31) $Q_{waste,y}$ = Quantity of fresh waste (dead animals) fed into incinerator in year y (t) $FF_{COM,c,y}$ = Fraction of fossil-based carbon in waste fed into incinerator in year y (t C/t). *Source: https://4vultures.org/wp-content/uploads/2023/10/Mitigating-GHG Emissions_vultures_PlazaandLambertucci2022.pdf* $FFC_{waste,y}$ = Combustion efficiency of incinerator c in year y (fraction). *Source: Chapter 5, Volume 5 (Incineration and open burning of waste) of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, page 5.20, para 5.4.1.3; —For waste incinerators it is assumed that the combustion efficiencies are close to 100 percent* Different categories of dead animals (eg: pigs, cows, dogs, etc) are brought at the site every day. A log of the count of no of animals of each category is maintained. There is no arrangement of weighing the waste, but it is made sure that the incinerator is operated at its maximum capacity i.e 250kg/hr. The data for number of working hours of the incinerator is obtained from the plant. The quantity of fresh waste is obtained by multiplying the number of dead animals to the capacity of incinerator and number of working hours, the calculations for which are as given below: $Q_{waste,y} = No.of\ waste\ animals\ incinerated \times No.of\ working\ hours\ of\ incinerator \times Capacity\ of\ incinerator$ Equation (32) Other than dead animals, incinerator also involves burning of RDF as fuel. Emissions due to RDF combustion can be calculated either using tool 3- Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion or equation (28) above. **Project emissions of CH_4 and N_2O from combustion within the project boundary ($PE_{COM,CH4,N2O,c,y}$)** $PE_{COM,CH4,N2O,y}^{c} = Q_{waste,c,y} \times (EF_{N2O,t} \times GWP_{N2O} + EF_{CH4,t} \times GWP_{CH4})$ Equation (33) Where: $PE_{COM,CH4,N2O,y}$ = Project emissions of CH_4 and N_2O from combustion within the project boundary of fossil carbon in combustor c in year y (t CO_2). $Q_{waste,c,y}$ = Quantity of fresh waste or RDF/SB fed into combustor c in year y (t). $EF_{N2O,t}$ = Emission factor for N_2O associated with waste treatment process t (t N_2O/twaste). $EF_{CH4,t}$ = Emission factor for CH_4 associated with treatment

process t (t CH4/t waste).
GWP_{N2O} = Global Warming Potential of nitrous oxide (t CO_2e/t N_2O)
GWP_{CH4} = Global Warming Potential of methane valid for the commitment period (tCO_2e/tCH_4)

Default values for this equation are:

Parameter	N_2O	CH_4
EF	1.21*60*10^-6	1.21*237*10^-3
GWP	265	28

Source: ACM0022, data and parameters not monitored, table 7 Thereby,

Project emissions due to incineration: 113194 t CO_2 e.

Plastic to Fuel (PtoF):

The pyrolysis plastic-to-fuel process is a chemical process that converts waste plastic into usable fuels, such as diesel, gasoline, etc. In this process, plastics are heated in absence of oxygen to temperatures typically between 600–800°C. This high heat causes the plastic polymers to break down into smaller molecules, producing a mixture of liquid hydrocarbons (pyro oil), pyro gas, and char (a solid residue). Pyrolysis helps reduce plastic waste and provides an alternative energy source, making it an attractive option for waste management and sustainable energy production.
In this process since the plastic is burned in absence of oxygen, there are no carbon emissions. However, the fuel required for heating can emit carbon. But in the present case, pyro-gas is recycled and used for heating the plastic. Another product, pyro-oil, is combusted in incineration causing carbon emissions. The by-product char is stored.
The plastic-to-fuel process employed in this plant cannot be classified as gasification, as the process operates under pyrolysis, which involves the thermal decomposition of plastic waste in the absence of oxygen, unlike gasification, which requires partial oxidation. As mentioned before, the pyrolysis gas (pyro-gas) is partly converted into oil, while a portion is reused as a heat source for processing the feed material, thus reducing external energy requirements. The remaining pyro-gas is stored in a gas balloon. Since the pyro-gas is either utilized within the process or stored, and there is no direct release of emissions, the net emissions from the plastic-to-fuel process are considered zero.
There are no leakage emissions in plastic to fuel process.

Briquetting:

Horticulture waste is converted to briquettes by shredding and compacting. This process consumes electricity. The equation 28, accounts for these emissions. The product from this process i.e briquettes are sold outside the project boundary.
There are no project emissions.

Leakage emissions:

Composting (COMP)
Leakage emissions from composting ($LE_{COMP,y}$) shall be accounted for if compost is subjected to anaerobic storage or disposed of in a SWDS.
In this case, the compost is sun dried and then packed into bags for selling purpose. Hence $LE_{COMP,y}$ is zero.

Leachate biomethanation (LB):

	The leakage emissions associated with the anaerobic digester (LE,) depend on how the digestate is managed. They include emissions as: $LE_{AD,y} = LE_{storage,y} + LE_{COM,y}$ Equation (34) Where, $LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO_2e) $LE_{COM.y}$ = Leakage emissions associated with composting of digestate ($LE_{COMP,y}$). **Leakage emissions associated with storage of digestate in year y (t CO_2e)** $LE_{storage,y}$ is applied in the case that the digestate is stored under the following anaerobic conditions: (a) In an un-aerated lagoon that has a depth of more than one meter; or (b) In a SWDS, including stockpiles that are considered a SWDS as per the definitions section. Storage of digestate under anaerobic conditions can cause CH_4 emissions due to further anaerobic digestion of the residual biodegradable organic matter. The procedure for determining $LE_{storage,y,}$ is distinguished for liquid digestate and solid digestate. In the present case, digestate is in liquid state, emissions for which can be calculated using following equation: $LE_{storage,y} = LE_{SD,CH4,default} \times Q_{CH4} \times GWP_{CH4}$ Equation (35) Where: $LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO_2e) $LE_{SD,CH4,default}$ = Default factor for the methane generation capacity of solid digestate (fraction) Q_{CH4} = Quantity of methane produced in the anaerobic digester in year y (t CH_4) GWP_{CH4} = Global warming potential of CH_4 (t CO_2/t CH_4) **Leakage emissions associated with composting of digestate ($LE_{COMP,y}$).** The digestate is not composted in the present case. Hence, $LE_{COMP,y}$ will be 0. Leakage emissions are amounted to be 176 tCO_2. **RDF manufacturing (RDF):** Leakage emissions: RDF manufacturing has leakage emissions which are calculated as given below: Leakage emissions associated with RDF/SB are accounted for the organic waste by products of the treatment process (not by-products from the RDF/SB combustor), which may be composted or disposed of in a SWDS, and the end-use of RDF/SB that is exported. The leakage emissions from RDF manufacturing are calculated by the following equation: $LE_{RDF_{SB},y} = LE_{ENDUSE,RDF\_SB} + LE_{SWDS,WBP_{RDF\ SB},y}$ Equation (36) Where:

$LE_{RDF\_SB,y}$ = Leakage emissions associated with RDF/SB in year y (t CO_2e)
$LE_{ENDUSE,RDF\_SB,y}$ = Leakage emissions associated with the end-use of RDF/SB
$LE_{SWDS,RDF\_SB,y}$ = Leakage emissions associated with disposing of waste by products associated with RDF/SB production in a SWDS in year y (t CO_2e).

The leakage emissions associated with disposing of waste by products associated with RDF/SB production in a SWDS in year y (t CO_2e) is considered to be zero as in the present case, the organic waste associated with RDF are sent for composting and not sent to the SWDS.

Leakage emissions associated with the end-use of RDF/SB that is exported outside the project boundary in year y are that it may be combusted or decomposed anaerobically.

In the present case, the RDF/SB exported in year y has end use as a fuel that is combusted. Documented evidence is provided that the RDF/SB exported off-site is combusted. Leakage emissions $LE_{ENDUSE,RDF\_SB,y}$ shall be calculated, according to the procedure below;

Leakage emissions from combusted off-site end use of RDF/SB ($LE_{ENDUSE,RDF\_SB,y}$):

$$LE_{ENDUSE_{RDF_{SB}},y} = Q_{RDF_{SB},COM,y} \times NCV_{RDF,y} \times EF_{CO2,RDF_{SB},y}$$

Equation (37)

Where:
$LE_{ENDUSE,RDF\_SB,y}$ = Leakage emissions of CO_2 from off-site combustion of RDF/SB in year y (tCO_2)
$Q_{RDF\_SB,COM,y}$ = Quantity of RDF/SB exported off-site with potential to be combusted in year y (t)
$EF_{CO_2,y}$ = CO_2 emissions factor for RDF/SB in year y (t CO_2/GJ)
IPCC guidelines, Chapter1-Introduction-table 1.4 - Municipal waste (non biomass fraction)
$NCV_{RDF\_SB,y}$= Net calorific value of RDF in year y (TJ/Gg)
IPCC guidelines, Chapter1-Introduction-table 1.2 – Municipal waste (non biomass fraction)

The emissions due to combustion of RDF sold outside the project boundary are as follows:

Amount of coal replaced by the RDF produced is calculated as:

$M_{coal} = M_{RDF} *(NCV_{RDF}/NCV_{coal})$

Leakage emissions are amounted to be 239327 t CO_2.

Emissions reduction:

Hence emission reduction is calculated:

$$Emissions\ reduction = Baseline\ emissions - Project\ emissions - Leakage\ emissions$$

Year	BE	PE	LE
2017	233,058	14,859	18,414
2018	136,102	21,686	31,993
2019	159,773	28,564	33,294

	<table><tr><td>**2020**</td><td>166,442</td><td>30,287</td><td>30,458</td></tr><tr><td>**2021**</td><td>209,958</td><td>33,831</td><td>31,568</td></tr><tr><td>**2022**</td><td>216,615</td><td>31,838</td><td>33,190</td></tr><tr><td>**2023**</td><td>252,825</td><td>32,268</td><td>37,757</td></tr><tr><td>**2024**</td><td>193,410</td><td>36,130</td><td>22,823</td></tr><tr><td>**Total**</td><td>1,568,181</td><td>229,463</td><td>239,497</td></tr></table> ER = 1,568,181-229,463-239,497=1,099,221 tCO_2e. As per calculations, with help of various project activities to treat the daily MSW, emissions that have been reduced for a period of 7 years from 2017 to 2024: 1,099,221 tCO_2 e. This activity reduces GHGs to the amount mentioned and hence reduces the climate change risk.
Findings	No findings were raised.
Conclusion	The verification team confirms that the estimation of emission reductions has been carried out correctly in accordance with the approved methodology, UCR requirements, and the approved PCN.

(.a.vi) Monitoring Report

<table><tr><td rowspan="2">Means of Project Verification</td><td>The monitoring contains the following parameters as required:</td></tr><tr><td><table><tr><td>Sl. No:</td><td>Parameter</td><td>Description</td></tr><tr><td colspan="3">ACM0022, version 3.0</td></tr><tr><td>1.</td><td>W_x or W_i
Total amount of waste disposed in a SWDS in year x</td><td>The total amount of waste disposed is measured on a wet-weight basis and monitored using the weighbridge as each truck enters the facility. Readings are recorded manually as well as digitally for every incoming truck, and these records along with the ER sheet were cross-checked by the verification team during the on-site visit. The annually summed data is provided below:
<table><tr><th>Year</th><th>Total MSW (tonnes)</th></tr><tr><td>2017</td><td>142397</td></tr><tr><td>2018</td><td>199279</td></tr><tr><td>2019</td><td>210099</td></tr><tr><td>2020</td><td>211952</td></tr><tr><td>2021</td><td>232014</td></tr><tr><td>2022</td><td>238638</td></tr><tr><td>2023</td><td>273912</td></tr><tr><td>2024</td><td>132228</td></tr></table>The verification team confirms that the data has been appropriately sourced, reviewed, and correctly applied for project emission calculations.</td></tr><tr><td>2.</td><td>$p_{n,j,x}$ or $p_{n,j,i}$
Weight fraction of the waste type j in the sample n collected during the year x or</td><td>The verification team has checked the waste characterization reports submitted by the PP and found that weight fraction has been checked on sample basis every three months. The same values have been used in calculations in the ER sheet and found to be correct.
PP is using quadrant method for sampling which is found to be correct.</td></tr></table></td></tr></table>

<table>
<tr><td></td><td>month i</td><td></td></tr>
<tr><td>3.</td><td>z_x
Number of samples collected during the year x</td><td>The parameter is measured in number and minimum three samples every four months are being collected during the year.
The verification team has checked the waste characterization reports and confirm that the values used in ER sheet are correct and used correctly.</td></tr>
<tr><td>4.</td><td>$Q_{waste,c,y}$
Quantity of fresh waste or RDF/SB fed into combustor c in year y</td><td>The parameter is measured using calibrated weigh scales with continuous monitoring, as described in the monitoring plan in the MR. The verification team reviewed the ER sheet and RDF records, daily feeding logs, and pre-sorting procedures, and confirms that the measurement process is consistent with the methodology requirements and the QA/QC procedures outlined in the MR.
Value applied: Measured with calibrated scales as given below:
<table>
<tr><th>year</th><th>RDF combusted (Qwaste,c,y)</th></tr>
<tr><td>2017</td><td>5335</td></tr>
<tr><td>2018</td><td>6749</td></tr>
<tr><td>2019</td><td>9519</td></tr>
<tr><td>2020</td><td>9311</td></tr>
<tr><td>2021</td><td>11884</td></tr>
<tr><td>2022</td><td>7981</td></tr>
<tr><td>2023</td><td>8384</td></tr>
<tr><td>2024</td><td>4396</td></tr>
</table>
The verification team confirms that the parameter has been appropriately monitored, accurately reported, and correctly applied in calculating project emissions from combustion within the project boundary based on the supporting documentation and onsite cross-checks.</td></tr>
<tr><td>5.</td><td>$EC_{t,y}$
Electricity consumption from the grid</td><td>The parameter is monitored continuously using calibrated energy meters, with meter nos. 065-04576712 and 24005723, and is recorded monthly. The verification team reviewed monthly electricity bills, calibration certificates, the ER sheet, and the MR summaries, and confirms that the monitoring approach is consistent with the monitoring plan and compliant with TOOL05 requirements.
Annual consumption values are provided below:
<table>
<tr><th>Year</th><th>Total electricity consumed (MWH)</th></tr>
<tr><td>2017</td><td>355</td></tr>
<tr><td>2018</td><td>575</td></tr>
<tr><td>2019</td><td>473</td></tr>
<tr><td>2020</td><td>483</td></tr>
<tr><td>2021</td><td>481</td></tr>
<tr><td>2022</td><td>445</td></tr>
<tr><td>2023</td><td>697</td></tr>
<tr><td>2024</td><td>726</td></tr>
</table>
The verification team confirms that the parameter</td></tr>
</table>

			has been correctly monitored, accurately reported, and appropriately applied for project emissions from electricity consumption within the project boundary.
	6.	***EC_{BE}*** Electricity generated by the Biogas generator in Leachate bio-methanation processes used for heating purpose in the incinerator.	The parameter is monitored through the electricity generation meter installed at the facility and records electricity generation (in MWh) by the biogas generator during leachate bio-methanation, used for heating in the incinerator. The verification team reviewed the electricity generation records and the ER sheet and confirms that the monitoring plan and methodology requirements have been accurately applied. As part of QA/QC, the electricity meter undergoes regular maintenance and testing as per supplier specifications to ensure measurement accuracy. Annual electricity generation values are as follows:

Year	Electricity generated - kWh
2017	22465
2018	41862
2019	49216
2020	45714
2021	45811
2022	47991
2023	48190
2024	33948

The verification team confirms that the parameter has been correctly monitored, reliably recorded, and appropriately applied for emission calculations related to project electricity generation within the project boundary.

	7.	***$Q_{RDF\_SB,COM,y}$*** Quantity of RDF/SB exported off-site with potential to be combusted in year *y*	The parameter is measured using sale invoices with monitoring while RDF trucks are leaving, as described in the monitoring plan in the MR. The verification team reviewed the ER sheet and RDF records, and weight records, and confirms that the measurement process is consistent with the methodology requirements and the QA/QC procedures outlined in the MR. **Value applied:** Measured with calibrated scales as given below:

year	RDF exported (tonnes) ($Q_{RDF\_SB,COM,y}$)
2017	20,069
2018	34,866
2019	36,281
2020	33,189
2021	34,399
2022	36,167
2023	41,148
2024	24,869

The verification team confirms that the parameter

			has been appropriately monitored, accurately reported, and correctly applied in calculating project emissions from combustion within the project boundary based on the supporting documentation and onsite cross-checks.
	8.	***$Q_{RDF\_SB,y}$*** Quantity of RDF/SB produced in year *y*	The parameter is measured using calibrated weigh scales with continuous monitoring, as described in the monitoring plan in the MR. The verification team reviewed the ER sheet and RDF records, daily feeding logs, and pre-sorting procedures, and confirms that the measurement process is consistent with the methodology requirements and the QA/QC procedures outlined in the MR. **Value applied:** Measured with calibrated scales as given below: <table><tr><th>year</th><th>RDF produced (tonnes) ($Q_{RDF\_SB,y}$)</th></tr><tr><td>2017</td><td>25,403</td></tr><tr><td>2018</td><td>41,615</td></tr><tr><td>2019</td><td>45,800</td></tr><tr><td>2020</td><td>42,501</td></tr><tr><td>2021</td><td>46,284</td></tr><tr><td>2022</td><td>44,148</td></tr><tr><td>2023</td><td>49,532</td></tr><tr><td>2024</td><td>29,266</td></tr></table>The verification team confirms that the parameter has been appropriately monitored, accurately reported, and correctly applied in calculating project emissions from combustion within the project boundary based on the supporting documentation and onsite cross-checks.
	9.	***$Q_{ww,y}$*** Amount of wastewater discharge generated by the project activity and treated anaerobically or released untreated from the project activity in year *y*	The parameter is measured using volume of tank, as described in the monitoring plan in the MR. The verification team reviewed the ER sheet and, wastewater discharge reports and confirms that the measurement process is consistent with the methodology requirements and the QA/QC procedures outlined in the MR. **Value applied:** the values applied in ER calculation are given below: <table><tr><th>Year</th><th>Leachate (m^3) ($Q_{ww,y}$)</th></tr><tr><td>2017</td><td>8030</td></tr><tr><td>2018</td><td>13933</td></tr><tr><td>2019</td><td>15954</td></tr><tr><td>2020</td><td>16219</td></tr><tr><td>2021</td><td>16557</td></tr><tr><td>2022</td><td>16875</td></tr><tr><td>2023</td><td>16490</td></tr><tr><td>2024</td><td>11140</td></tr></table>The verification team confirms that the parameter has been appropriately monitored, accurately reported, and correctly applied in calculating project emissions from combustion within the project boundary based on the supporting

<table>
<tr><td></td><td></td><td>documentation and onsite cross-checks.</td></tr>
<tr><td>10.</td><td>$COD_{BL,y}$
COD of the wastewater discharge generated by the project activity in year y</td><td>The COD of wastewater discharge has been measured through internal lab analysis once a week. The values applied in ER sheet are based on the average of the values achieved during the lab analysis and found to be correct.
The verification team has checked the values achieved and the analysis done by the external agency and found to be inline with the applied methodology and found to be correct.</td></tr>
<tr><td>11.</td><td>Qbiogas,y
Amount of biogas collected at the digester outlet in year y</td><td>The parameter is measured using volumetric flow measurement, the gas is collected in a balloon which has a tie rope indicator based on which the volume is recorded, as described in the monitoring plan in the MR. The verification team reviewed the ER sheet and, indicator values and confirms that the measurement process is consistent with the methodology requirements and the QA/QC procedures outlined in the MR.
Value applied: the values applied in ER calculation are given below:
<table>
<tr><th>Year</th><th>Biogas produced (m^3) ($Q_{biogas,y}$)</th></tr>
<tr><td>2017</td><td>14</td></tr>
<tr><td>2018</td><td>27</td></tr>
<tr><td>2019</td><td>31</td></tr>
<tr><td>2020</td><td>30</td></tr>
<tr><td>2021</td><td>30</td></tr>
<tr><td>2022</td><td>31</td></tr>
<tr><td>2023</td><td>31</td></tr>
<tr><td>2024</td><td>22</td></tr>
</table>
The verification team confirms that the parameter has been appropriately monitored, accurately reported, and correctly applied in calculating project emissions from combustion within the project boundary based on the supporting documentation and onsite cross-checks</td></tr>
<tr><td colspan="3">AMS I.D. version 18.0</td></tr>
<tr><td>12.</td><td>$EG_{PJ,facility,y}$
Net electricity supplied to the grid facility by the project activity.</td><td>The parameter is continuously measured using a bidirectional energy meter, with meter serial number 24005723, with monthly recordings archived electronically through the Surya Logix mobile app and summarized annually. The meter, installed outside the project boundary, captures both import from and export to the grid, while the Surya Logix control system monitors all electricity generation, supply, and consumption to ensure accurate and transparent data management.
The verification team cross-checked the data during the desk review and onsite visit at NWMPL and confirms that the values have been correctly recorded. Calibration certificates were reviewed and confirm that the meter is properly calibrated in accordance with technical requirements.</td></tr>
</table>

The verification team confirms that the monitoring report and monitored parameters are in line with the approved PCN and the applied

	methodology.
Findings	CL 03, CL 04, CL 05 and CAR 03 were raised and closed successfully. Refer to “Clarification request, corrective action request and forward action request” below for further details.
Conclusion	The verification team confirms that the data and parameters have been monitored and reported in accordance with the approved methodology and PCN, and that the emission reduction calculations are accurate and consistent with UCR requirements.

Start date, crediting period and duration

Means of Project Verification	The verification team confirms that the project has a crediting period of 7 years and 11 months, starting from 15/02/2017 to 31/12/2024, both dates inclusive. This has been verified through a detailed desk review of the project documentation. During the onsite verification, it was observed that physical project activities commenced on 15/01/2017, while the first invoice was issued on 15/02/2017, which is considered the start date based on the first billing and is consistent with the declared start date. Based on these sources, the verification team confirms that the crediting period start date of 15/02/2017 is accurate and appropriate.
Findings	CAR 01 was raised and closed. Refer to “Clarification request, corrective action request and forward action request” below for further details.
Conclusion	The verification team confirms that the crediting period of 7 years and 11 months, from 15/02/2017 to 31/12/2024 (both dates inclusive), is valid and 15/02/2017 accurately reflects the start date of the project activity. Furthermore, the project start date complies with section 2.6 of program manual, version 6.1, and is consistent with the approved PCN.

Positive Environmental impacts

Means of Project Verification	The project mitigates methane emissions and leachate from conventional waste disposal, significantly reducing the need for new landfill sites and enabling better land use for housing or infrastructure. Optimized transportation routes lower emissions from municipal solid waste (MSW) movement, while electricity generation from RDF and solar panels replaces fossil fuel-based power, cutting greenhouse gases. Biomining of legacy waste reclaims land, reduces pollution, and recovers valuable materials. By diverting waste from open burning, air pollution and related health risks are minimised. Leachate collection and treatment systems prevent groundwater and surface water contamination, and segregation of hazardous materials safeguards soil quality. Composting organic waste produces nutrient-rich compost, enhancing soil fertility and reducing reliance on chemical fertilizers, thereby promoting sustainable environmental management. Horticulture waste is processed avoiding fire hazards. The project replaces open, unsightly, unsanitary landfills infested with pathogens to sanitary, pleasant landscapes.
Findings	No findings were raised.
Conclusion	The verification team confirms that the project is in technical conformance with UCR requirements and demonstrates verifiable environmental benefits, meeting the criteria for environmental impact verification under the UCR framework.

Project Owner- Identification and communication

Means of Project Verification	NWMPL has been identified as the project owner/participant, and all communication with the UCR has been conducted by NWMPL. This has been verified through email correspondence with UCR.
Findings	No findings were raised.
Conclusion	The verification team confirms that the project owner is in compliance

	with the requirements of the UCR program manual, version 6.1.

Positive Social Impact

Means of Project Verification	The project has demonstrated positive social impacts by improving the hygienic management of municipal solid waste, thereby enhancing public health and living conditions in the city. Manual and mechanical segregation of waste ensures the removal of non-biodegradable and hazardous materials, reducing exposure of rag pickers and the local population to serious health risks associated with unmanaged waste. Recyclable materials are either reused within the plant or sold through local contractors, providing direct monetary benefits and supporting local livelihoods. Additionally, the project generates both direct and indirect employment opportunities for the region, contributing to socioeconomic development.
Findings	No findings were raised.
Conclusion	The verification team confirms that the project activity has delivered positive social impacts by improving hygienic management of municipal solid waste, reducing health risks for rag pickers and the local population, generating direct and indirect employment, and supporting local livelihoods through recycling and reuse of materials.

Sustainable development aspects (if any)

Means of Project Verification	The verification team assessed the sustainable development claims in the MR through review of supporting documents, monitoring data, and site observations. Alignment with SDG Goals 2, 4, and 13 was confirmed based on evidence of pollution prevention, employment generation, skill development through MSWM operations, methane avoidance, biogas generation, solar energy integration, and resulting GHG emission reductions. Indirect contributions to SDGs 1, 3, 5, 6, 7, 8, 11, 12, 14, and 15 were also validated. Environmental and socioeconomic benefits were verified through groundwater protection measures, absence of open burning, scientific landfill development, biomining, recyclables recovery, composting, RDF production, leachate biomethanation, and circular-economy outputs. Consistency between the MR, operational logs, waste processing data, biogas and leachate treatment records, staffing data, invoices, and JMRs confirmed the accuracy of the claims and compliance with UCR requirements. No negative environmental or social impacts were observed, and the project demonstrates clear contributions to sustainable development.
Findings	No findings were raised.
Conclusion	The project directly contributes to SDG 2, 4, and 13 through improved sanitation, employment and skill development in MSWM operations, and measurable climate action benefits from methane avoidance, biogas generation, and renewable energy use. Indirect contributions to SDGs 1, 3, 5, 6, 7, 8, 11, 12, 14, and 15 were verified through enhanced hygiene, resource recovery, and reduced environmental contamination. The verification team confirms that the project complies with the UCR Standard Protocol, aligns with the UN SDGs, and delivers clear environmental, social, and economic benefits that complement its GHG mitigation outcomes.

Internal quality control

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The verification report prepared by the Team Leader is reviewed by an Independent Technical Reviewer, who is independent to the verification team and possesses the necessary competence in the relevant technical area. This review is conducted to ensure that the internal procedures established by KBS have been duly followed, and that the verification report and opinion have been reached objectively and in compliance with applicable UCR requirements.
The Independent Technical Reviewer may approve or reject the verification report. Findings may be identified during this stage, which must be satisfactorily resolved before the Request for Issuance is submitted to UCR. Once approved by the Manager – Technical & Certification, the final authorization is granted by the Managing Director, KBS.

Project Verification opinion

KBS Certification Services Ltd. has been contracted by Nashik Waste Management Pvt. Ltd. (NWMPL) to undertake the first independent verification and certification of greenhouse gas (GHG) emission reductions registered under UCR, titled “Integrated Municipal Solid Waste Management Activities Programme, Nashik, India, Nashik”, UCR Ref. No. 492. This verification covers the first monitoring period from 15/02/2017 to 31/12/2024 (both dates inclusive), within the overall crediting period from 15/02/2017 to 31/12/2024. Under UCR requirements, all registered projects must undergo independent third-party verification and certification of emission reductions as the basis for the issuance of Carbon Offset Units (COUs).
The verification has been conducted based on the approved Project Concept Note (PCN) and the Monitoring Report submitted for the project. Our verification approach follows the requirements set forth in the UCR Project Verification Standard.
The management of NWMPL is responsible for preparing the GHG emissions data and the reported GHG emission reductions, as outlined in the Final Monitoring Report. The calculation and determination of GHG emission reductions, as well as the development and maintenance of records and reporting procedures, are the responsibility of NWMPL in accordance with the Monitoring Report.
It is the responsibility of KBS Certification Services Ltd. to express an independent verification opinion on the reported GHG emissions and the calculation of GHG emission reductions for the monitoring period 15/02/2017 to 31/12/2024 (both dates inclusive), based on the data provided in Final Monitoring Report – Version 1.3.
Based on our understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate those risks, KBS planned and performed the verification to obtain sufficient information and explanations necessary to provide reasonable assurance that the reported GHG emission reductions for the period are fairly stated.
Verified and certified emission reductions reporting period: 15/02/2017 to 31/12/2024 (both dates inclusive).

	Amount	Unit
Baseline emissions (BE_y)	15,68,181	tCO_2e
Project emissions (PE_y)	2,29,463	tCO_2e
Leakage emissions (LE_y)	2,39,497	tCO_2e
Emission reductions (ERs)	10,99,221	tCO_2e

Abbreviations

Abbreviations	Full texts
BE	Baseline Emissions
BG	Biogas Generator
CARs	Corrective Action Request
CDM	Clean Development Mechanism
CDM-PDD	Clean Development Mechanism Project Design Document
CERs	Certified Emission Reductions
CH_4	Methane
CL	Clarification Request
CO_2	Carbon Dioxide
COD	Chemical Oxygen Demand
COUs	Carbon Offset Units
CPA	CDM Component Project Activity
DG	Diesel Generator
DOC	Degradable Organic Carbon
ERs	Emission Reductions
FARs	Forward Action Request
FOD	First-Order Decay
GHG	Green House Gases
GWP	Global Warming Potential
JMR	Joint Meter Reading
LB	Leachate Biomethanation
LCA	Life Cycle Analysis
LE	Leakage Emissions
Ltd.	Limited
MRF	Material Recovery Facility
MSW	Municipal Solid Waste
MSWM	Municipal Solid Waste Management
NCV	Net Calorific Value
NWMPL	Nashik Waste Management Pvt. Ltd.
PCN	Project Concept Note
PDD	Project Design Document
PE	Project Emissions
PoA	Programme Of Activities
PP	Project Participant
PV	Solar Photovoltaic
PVR	Project Verification Report
Pvt.	Private
RDF	Refuse-Derived Fuel
SB	Stabilized Biomass
SDGs	Sustainable Development Goals
SLF	Scientific Landfill
SW	Solid Waste
SWDS	Solid Waste Disposal Site
tCO_2e	Tonnes Of Carbon Dioxide Equivalent
TR	Technical Reviewer
UCR	Universal Carbon Registry
UNFCCC	United Nation Framework for Climate Change Convection

Competence of team members and technical reviewers

>>

Personnel Name	Mr. Rishabh Madan					
Schemes	☒ CDM	☒ GCC	☐ GS	☒VCS	☒A6.4	☒Other GHG Schemes (Cercarbono, UCR, ISO

						14064, ISO 14067, SDVista, JCM)

Qualified to work as			
Team Leader	☒	Technical Expert	☒
Validator/Verifier	☒	Financial Expert	☐
Technical Reviewer	☐	Local Expert (India)	☒

Area(s) of Technical Expertise	
Sectoral Scope	**Technical Area**
SS 1: Energy industries (renewable/non-renewable sources	TA 1.2. Renewables
SS 7: Transport	TA 7.1. Transport
SS 10: Fugitive emissions from fuels (solid, oil and gas)	TA 10.1. Fugitive emissions from oil and gas
Approved by (Manager C &T)	Mr. Dushyant Parashar
Approval date	27/12/2024

Personnel Name	Mr. Ashish Yadav					
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☒A6.4	☒Other GHG Schemes (Cercarbono, SDvista, VCS CCB)

Qualified to work as			
Team Leader	☒	Technical Expert	☒
Validator/Verifier	☒	Financial Expert	☐
Technical Reviewer	☒	Local Expert (India)	☒

Area(s) of Technical Expertise	
Sectoral Scope	**Technical Area**
SS: 1 Energy Industries (Renewable/non-renewable)	TA 1.2. Renewables
SS: 3 Energy demand	TA 3.1 Energy demand
SS 13: Waste handling and disposal	TA 13.1. Solid waste and wastewater
Approved by (Manager C&T)	Mr. Dushyant Parashar
Approval date	28-10-2024

Personnel Name	Ms. Alen Mariyam Thomas					
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☒ A6.4	☒Other GHG Schemes (SDvista, CCB)

Qualified to work as			
Team Leader	☐	Technical Expert	☐
Validator/Verifier	☒	Financial Expert	☐
Technical Reviewer	☐	Local Expert (India)	☒

Area(s) of Technical Expertise	
Sectoral Scope	**Technical Area**
-	-
Approved by (Manager Competence & Training)	Mr. Dushyant Parashar
Approval date	29/11/2024

Personnel Name	Mr. Praveen N Urs				
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☒Other GHG Schemes (VCS CCB, ICR,

			CERCARBONO, Social Carbon, ISO 14064)
Qualified to work as			
Team Leader	☒	Technical Expert	☒
Validator/Verifier	☒	Financial Expert	☒
Technical Reviewer	☒	Local Expert (India)	☒
Area(s) of Technical Expertise			
Sectoral Scope		**Technical Area**	
SS 1: Energy industries (renewable/non-renewable sources)		TA 1.1: Thermal energy generation	
		TA 1.2: Renewable	
SS 13: Waste handling and disposal		TA 13.1: Solid waste and wastewater	
		TA 13.2: Manure	
SS- 14: Afforestation and Reforestation		TA- 14.1: Afforestation and Reforestation	
SS – 15 – Agriculture		TA 15.1: Agriculture	
Approved by (Manager Competence & Training)		Mr. Dushyant Parashar	
Approval date		12-05-2024	

Document reviewed or referenced

No.	Author	Title	References to the document	Provider
1.	Project Proponent	Project Concept Note	Final v1.2 dt. 07/11/2025	PP
2.	Project Proponent	Monitoring report	Initial v1.0 dt. 09/06/2025 Final v1.3 dt. 22/04/2026	PP
3.	Project Proponent	ER spreadsheet	Final v1.2 dt. 08/11/2025	PP
4.	UCR	UCR CoU Standard Ver 7.0 UCR Verification Standard Ver2.0 UCR Program manual Version 6.1	https://www.ucarbonregistry.io/Document?projectCategoryId=1	UCR
5.	UNFCCC	CDM Methodologies ACM0022: Large-scale: Consolidated Methodology: Alternative waste treatment processes, Version 03.0 AMS-I. D: Small-scale Methodology: Grid connected renewable electricity generation, Version 18.0		UNFCCC
6.	Project Proponent	Waste Characterization reports	-	PP
7.	Project Proponent	Internal Lab records	-	PP
8.	Project Proponent	External lab records	-	PP
9.	Project Proponent	Weigh Bridge records	-	PP
10.	Project Proponent	Weigh bridge calibration records	-	PP
11.	MSEB	Electricity bills	-	PP
12.	Project Proponent	Sales invoices for RDF	-	PP
13.	Project Proponent	Wastewater discharge reports	-	PP
14.	Project Proponent	Biogas produced flow indicators	-	PP
15.	Surya Logix	Solar plant control systems	-	PP
16.	IPCC	2006 IPCC Guidelines for National GHG Inventories Chapter 3, Volume 2	-	IPCC

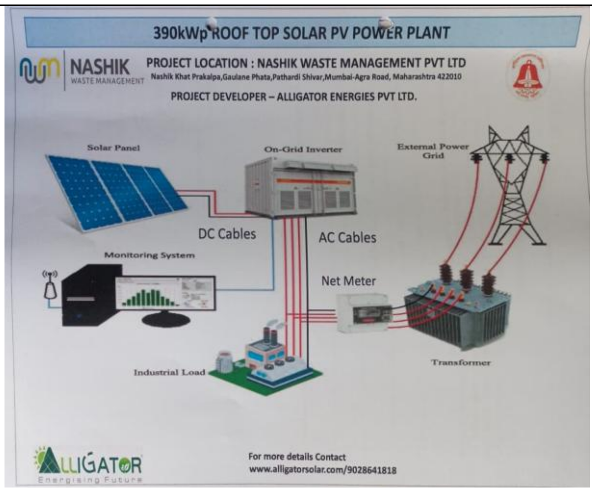
Clarification request, corrective action request and forward action request

Table 1. CLs from this Project Verification

<table>
<tr><th>CL ID</th><th>01</th><th>Section no.</th><th></th><th>Date: 19/08/2025</th></tr>
<tr><td colspan="5">Description of CL</td></tr>
<tr><td colspan="5">PP to provide clarification:
1. In the Single Line Diagram document, the serial number mentioned for the solar meter is X23138735. However, during the site visit, the meter observed for the solar panel had the serial number 24005723.
2. Further, the Single Line Diagram document mentions “Alligator Energies Private Limited” as both the Project Developer and the Customer Name. PP is requested to explain the role of Alligator Energies Private Limited in this project activity.
3. Pages 3–5 of the Single Line Diagram document mention the project as “VITP Private Limited”.</td></tr>
<tr><td colspan="4">Project Owner’s response</td><td>Date: 29/08/2025</td></tr>
<tr><td colspan="5">1. The differentiation between Net Meter, ABT Meter & Solar generation meter is provided in a tabular format, the differences between 3 meters, their Serial no., location and their exact role in the Solar unit is presented in the following table:</td></tr>
</table>

ABT NET METER	SR NO-24005723	A solar net meter is a bi-directional meter that measures both the electricity your solar panels send to the grid and the electricity you import from the grid. A SOLAR ABT NET METER refers to this type of bi-directional net meter that also incorporates or operates within an Availability-Based Tariff (ABT) system. In essence, the net meter tracks your solar energy contributions and consumption, while the "ABT" designation indicates it's used in a system where electricity pricing and charging are based on the real-time availability of power supply.	
GENERATION METER	SR NO-X2246958	This meter is installed to measure the gross amount of electricity generated by your solar PV system.	
ACDB MFM METER	SR NO-X2318735	In the context of solar energy, "MFM" most commonly refers to a Multi-Function Meter, a sophisticated digital device for measuring and monitoring various electrical parameters like voltage, current, power, and energy, essential for tracking solar system performance and managing energy consumption	

2. In the Roof Top Solar PV Power Plant at NWMPL, Growatt from China, the manufacturers of grid solar inverters have supplied the solar inverters through their distributors Growatt New Energy Technology Co., Ltd/official distributor of Growatt, to Alligator Energies Pvt, Ltd., of India, who in turn have installed the solar plant at NWMPL using these inverters; this is the supply chain. Hence you see both the Chinese Growatt and the Indian Alligator energies reflected in the evidence provided in the single line diagram document. In the first part of the supply chain Alligator is the customer, in the second part Alligator is the project developer.



Growatt

Growatt Warranty certificate

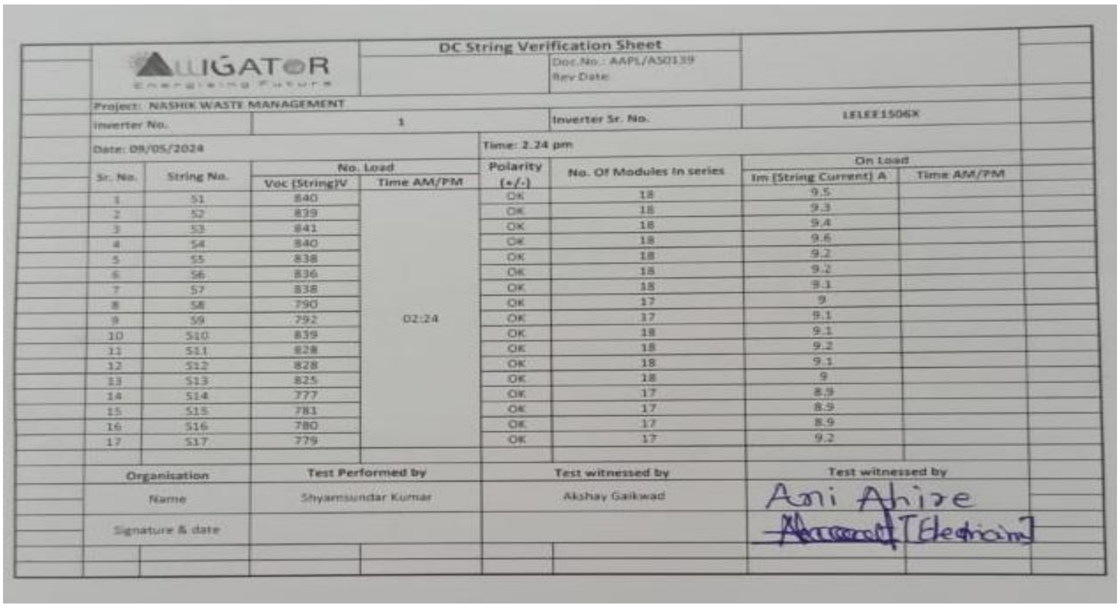
To Whom It May Concern

We, **Shenzhen Growatt New Energy Technology Co., Ltd.,** who is proven and reputable manufacturer of on grid solar inverters since 2010, having factory at #28, Guangming Road, Shiyan, Bao'an District, Shenzhen, China, hereby confirms warranty valid for **10 years** on Inverter **Serial Nos(SN)** as per the online link: https://www.ginverter.com/list-19.html, from the **date of purchase on the invoice** from **Growatt New Energy Technology Co., Ltd/official distributor of Growatt**

DISTRIBUTOR INFO	Photovoltaic Solar
DATE OF INVOICE-INVOICE No.	23/03/2024| PVS/23-24/2230
CUSTOMER NAME	Alligator Energies Private Limited

*For Growatt Warranty Terms and Conditions refer the Warranty Card provided with the inverter box at the time of purchase.

3. Project Name was incorrect in the string sheets provided by the Alligator Energies to NWMPL, they have been corrected since by the company and are shared here. Submitting the solar inverter string performance report carried out by the supplier Alligator Energies during the commissioning of solar plant at NWMPL site:



ALLIGATOR
Energising Future

DC String Verification Sheet

Doc.No.: AAPL/AS0139
Rev Date:

Project: NASHIK WASTE MANAGEMENT

Inverter No.	1	Inverter Sr. No.	LELEE1506X
Date: 09/05/2024		Time: 2.24 pm	

Sr. No.	String No.	No. Load		Polarity (+/-)	No. Of Modules in series	On Load	
		Voc (String)V	Time AM/PM			Im (String Current) A	Time AM/PM
1	S1	840	02:24	OK	18	9.5	
2	S2	839		OK	18	9.3	
3	S3	841		OK	16	9.4	
4	S4	840		OK	18	9.6	
5	S5	838		OK	18	9.2	
6	S6	836		OK	18	9.2	
7	S7	838		OK	18	9.1	
8	S8	790		OK	17	9	
9	S9	792		OK	17	9.1	
10	S10	839		OK	18	9.1	
11	S11	828		OK	18	9.2	
12	S12	828		OK	18	9.1	
13	S13	825		OK	18	9	
14	S14	777		OK	17	8.9	
15	S15	781		OK	17	8.9	
16	S16	780		OK	17	8.9	
17	S17	779		OK	17	9.2	

Organisation	Test Performed by	Test witnessed by	Test witnessed by
Name	Shyamsundar Kumar	Akshay Gaikwad	Ani Ahire
Signature & date			[Electrician]

ALLIGATOR
Energising Future

DC String Verification Sheet

Doc.No.: AAPL/AS0139
Rev Date:

Project: NASHIK WASTE MANAGEMENT

Inverter No.	2	Inverter Sr. No.	LELEE1S05J

Date: 09/05/2024 | Time: 2.24 pm

Sr. No.	String No.	No. Load		Polarity (+/-)	No. Of Modules In series	On Load	
		Voc (String)V	Time AM/PM			Im (String Current) A	Time AM/PM
1	S1	785	03:01	OK	17	9.5	
2	S2	785		OK	17	9.3	
3	S3	782		OK	17	9.4	
4	S4	782		OK	17	9.6	
5	S5	782		OK	17	9.2	
6	S6	783		OK	17	9.2	
7	S7	780		OK	17	9.1	
8	S8	781		OK	17	9	
9	S9	784		OK	17	9.1	
10	S10	782		OK	17	9.1	
11	S11	783		OK	17	9.2	
12	S12	787		OK	17	9.1	
13	S13	784		OK	17	9	
14	S14	779		OK	17	8.9	
15	S15	784		OK	17	8.9	
16	S16	780		OK	17	8.9	
17	S17	779		OK	17	9.2	

Organisation	Test Performed by	Test witnessed by	Test witnessed by
Name	Shyamsundar Kumar	Akshay Gaikwad	Anil Ahire
Signature & date			[Electrician]

ALLIGATOR
Energising Future

DC String Verification Sheet

Doc.No.: AAPL/AS0139
Rev Date:

Project: NASHIK WASTE MANAGEMENT

Inverter No.	3	Inverter Sr. No.	ADNME1900E

Date: 09/05/2024 | Time: 2.24 pm

Sr. No.	String No.	No. Load		Polarity (+/-)	No. Of Modules In series	On Load	
		Voc (String)V	Time AM/PM			Im (String Current) A	Time AM/PM
1	S1	769	02:24	OK	17	9.5	
2	S2	769		OK	17	9.3	
3	S3	768		OK	17	9.4	
4	S4	772		OK	17	9.6	
5	S5	768		OK	17	9.2	
6	S6	815		OK	18	9.2	
7	S7	815		OK	18	9.1	

Organisation	Test Performed by	Test witnessed by	Test witnessed by
Name	Shyamsundar Kumar	Akshay Gaikwad	Anil Ahire
Signature & date			[Electrician]

Documentation provided by Project Owner

1. *Evidence document shared in the drive: point 20. Instrumentation Specifications, serial numbers, and calibration records for all metering devices.*
2. *Evidence document shared in the drive: point 2. Single Line Diagram – Solar*
3. *Evidence is shared within this document above.*

UCR Project Verifier assessment	**Date:** 24/09/2025

1. PP has provided details of three meters: Net Meter, ABT Meter, and Solar Generation Meter. However, the specific location and exact role of each meter in relation to the solar system are not clearly defined. The MR also lacks information on meter numbers X2246958 and X2318735. Only Solar Generation Meter 24005723 is mentioned, along with meter 065-04576712, for which no supporting documents have been provided. Additionally, while ABT Meter is listed as 24005723 in the table, the folder contains documents for meter 24051021, which is addressed separately under CL 03. PP to provide supporting documents and revise the MR accordingly. **Hence, CL 01.1. is open.**
2. PP has clarified that Alligator Energies Pvt. Ltd. procured inverters from Growatt (China) as a customer and subsequently acted as the project developer for installation at NWMPL. Hence, its dual role is reflected in the Single Line Diagram. **Hence, CL 01.2. is closed.**
3. PP to revise the pages 3-5 of SLD and reshare the DC String Verification Sheet in the folder 2. **Hence, CL 01.3. is open.**

Project Owner's response	**Date:** 01/10/2025

CL 01.1.

a. Net Meter or the Generation meter **(No. X2246958)** is installed in the **Solar Control Unit** in the Electrical Panel Room and it measures the actual Solar units generation by the Solar Power System.

b. ABT Meter **(No. 24005723)** is the Bi-directional meter installed in the **Main Transformer Room** by MSEDCL (State Electricity Board), which gives the exact electrical units consumed by the Consumer (NWMPL) after export and import of the electricity to/from the Grid. The Meter 24005723 is a newly installed Electrical meter (after start of Net metering at NWMPL site in Sept 24 by MSEDCL, post solar unit installation) and the Sr No. is mentioned in the Oct 24 month Bill. This is the supporting document:

MAHARASHTRA STATE ELECTRICITY DISTRIBUTION CO. LTD.

MAHAVITARAN
Maharashtra State Electricity Distribution Co. Ltd.

BILL OF SUPPLY FOR THE MONTH OF OCT-2024 202410459584691

GSTIN:27AAECM2933K1ZB Website:www.mahadiscom.in HSN CODE:27160000

NASIK CIRCLE - 595 NASIK URBAN DIV II - 041 DWARKA SUB-DIVISION - 670 1

Consumer No. : 049139016010
Consumer Name : THE COMMISSIONER NMC
Address : RAJIV GANDHI BHAVAN
SHARANPUR ROAD
Village : NASIK **Pin Code :** 422010
Mobile No. : 99******33
Email : ***mech@nmc.gov.in

BILL DATE	08/11/2024	6,03,850
DUE DATE	22/11/2024	
IF PAID UPTO	14/11/2024	5,89,420
IF PAID AFTER	22/11/2024	6,11,400
Last Receipt No./Date	: 0000197864 / 09-10-2024	
Last Month Payment	: 6,54,700.00	

Security Deposit Held Rs. :	11,01,000.00	Addl. S.D. Demanded Rs. :	0.00
Bank Guarantee Rs. :	0	S.D. Arrears Rs. :	6,72,000.00

Details for making Energy Bill payment through RTGS/NEFT mode
o Beneficiary Name: MSEDCL o Beneficiary Account Number: **MSEDHT01049139016010**
o Name of Bank: State Bank of India o IFS Code: **SBIN0008965 (fifth,sixth and seventh character is zero)**
o Name of Branch: IFB, BKC Branch-MSEDCL
Disclaimer: Please use above bank details only for payment against consumer number mentioned in beneficiary account number.

GSTIN :	PAN : AAALN0085E	Metering Type : HT
Date of Connection : 26/06/2001	Tariff : 101 HT-I A	Meter No : SCHNEIDER(100)-24005723
Contract Demand (KVA) :300.00	Old Tariff : HT-I A	CT Ratio : 5/5
Connected Load (KW) : 1216.00	Elec. Duty : 06 PART B	PT Ratio : 11/110
Sanctioned Load (KW): 1216	Urban/Rural : Urban	Connected CT Ratio : 25/5
Feeder Voltage (KV) : 11	Seasonal :	Connected PT Ratio : 11/110
Feeder Name: 11KV VILHOLI CIDCO	Scale/Sector : Small Scale / Public Sector	
Express Feeder : No	Solar Conn./ Type: Yes /Net Metering	
Substation Name:33/11 KV CIDCO-II	Solar Capacity: 300	LIS Indicator :

c. The 3rd meter is CT – Current transformer **(No. 24051021)** which is **installed in the Solar Control Unit** in the Electrical Panel Room and it has actually no role in the electrical generation but is one of the accessories of the solar unit.

d. In addition to this another Meter mentioned in the document attached is the ACDB MFM meter, **(No. X2318735)** commonly referred to as Multi-function meter, which is a digital device for measuring and monitoring various electrical parameters like voltage, current, power and energy essential for tracking solar system performance and managing energy consumption. This is also installed in the Solar Control unit.

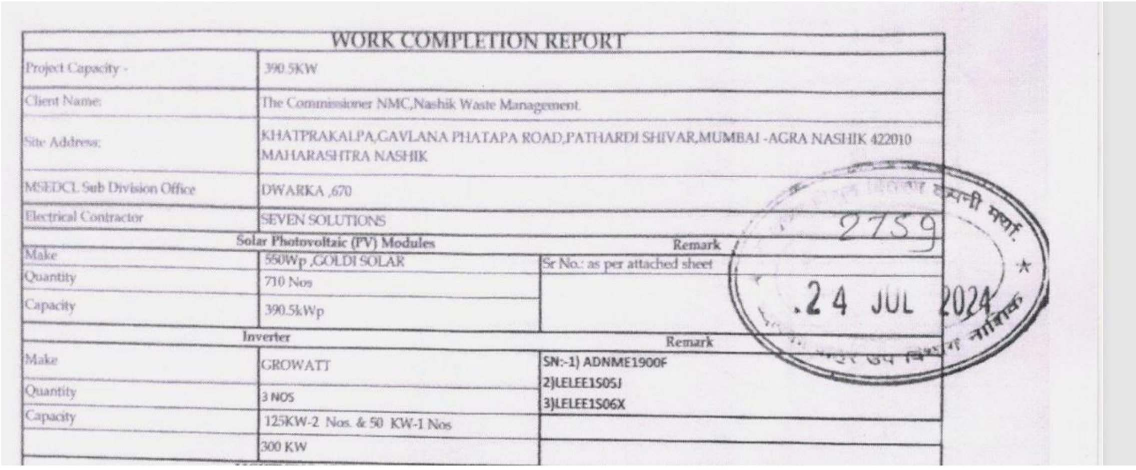
CL 01.3. Revised pages of DC String Verification Sheet (pg. 3-5) of SLD is reshared separately in the folder 2 of the drive.

Documentation provided by Project Owner	
Revised pages of DC String Verification Sheet (pg. 3-5) of SLD is reshared separately in the folder Single Line Diagram of Solar Power Generation Plant of the drive.	
UCR Project Verifier assessment	**Date:** 09/10/2025

CL 01.1. PP has provided detailed clarifications on the roles, meter numbers, and installation locations of all relevant meters associated with the solar power system. The Net Meter (X2246958) is identified as the generation meter located in the Solar Control Unit, the ABT Meter (24005723) is installed by MSEDCL in the Main Transformer Room for recording import and export of electricity, the Current Transformer meter (24051021) is confirmed to be an accessory with no direct role in generation, and the ACDB MFM Meter (X2318735) is a multifunctional monitoring device also located in the Solar Control Unit. Meters 24051021, X2246958, and X2318735 do not play a role in electricity generation, whereas ABT Meter 24005723 is responsible for recording import/export data. The previously missing meter numbers and their functions have been clarified and are found to be satisfactory. **Hence, CL 01.1. is closed.** **CL 01.3.** PP has now provided the revised DC String Verification Sheet in the Single Line Diagram of Solar Power Generation Plant and the same is accepted by the verification team. **Hence, CL 01.3. is closed.**

CL ID	02	**Section no.**		**Date:** 19/08/2025
Description of CL				
PP to provide clarification for: Capacity discrepancy: As per the letter dated 09-09-2024, the subject line reads "Final Release order for installation of solar net metering for connectivity/ installation of roof-top solar PV system of 300 kW (AC) in r/o M/s The Commissioner NMC, Compost Plant, Gaulane Road Pathardi, Nashik (Consumer No. 049139016010)," whereas the MR states the capacity as 0.39 MW, and 0.4 MW was observed during the on-site visit.				
Project Owner's response				**Date:** 29/08/2025
The connected electrical load to the NWMPL Processing Plant site at Nashik having Consumer No. 049084771715 is 300 Kw and so we had submitted a proposal to MSEDCL Nashik for a Solar Power System of 300 Kw electrical generation capacity. The electricity generated from Solar energy is in DC current and there are about 10-15 % losses while converting from DC to AC current. In order to make up for the conversion losses and also have electrical generation on a higher side, we have installed about 30 % higher capacity Solar Generation Panel system and BOS. Thus, the application submitted to MSEDCL (Maharashtra State Electricity Distribution Company Ltd.) by NWMPL is of 300 Kw (0.3 MW) AC Power capacity, while our Solar generation system is of 390 Kw (0.39 MW) capacity.				
Documentation provided by Project Owner				
UCR Project Verifier assessment				**Date:** 24/09/2025
PP clarified that a 300 kW (AC) system was proposed to MSEDCL, with a 390 kW DC system installed to offset conversion losses. However, the explanation lacks supporting documents and does not address: • Capacity discrepancies (MR: 0.39 MW, MSEDCL letter: 300 kW, site: 0.4 MW) • Mismatch in consumer numbers (049139016010 vs. 049084771715) PP to submit relevant documentation. **Hence, CL 02 is open.**				
Project Owner's response				**Date:** 01/10/2025
CL 02 Solar photovoltaic modules i.e., Panels installed capacity is 390 kw for higher generation. The Inverter capacity for Conversion into DC is 300 kw. Further it is to be mentioned here that the electrical generation from Solar unit is in DC form and while its conversion in to AC form there is Conversion loss of about 10-15 %. So even in extreme bad weather or low illumination conditions in order to have at least 300 kw power generation with enough margin, we have installed the Solar panels capacity on higher side. Thus, the actual capacity of the Solar System is 300 Kw with AC Current.				

As seen in row no. 2 of the Work Completion Report below, Solar Photovoltaic Module capacity and row no. 3 Inverter capacity dated 26th July 2024 from Additional Executive Engineer, MSEDCL, the capacity issue is clarified.



WORK COMPLETION REPORT

Project Capacity -	390.5KW	
Client Name:	The Commissioner NMC,Nashik Waste Management.	
Site Address:	KHATPRAKALPA,GAVLANA PHATAPA ROAD,PATHARDI SHIVAR,MUMBAI -AGRA NASHIK 422010 MAHARASHTRA NASHIK	
MSEDCL Sub Division Office	DWARKA ,670	
Electrical Contractor	SEVEN SOLUTIONS	
Solar Photovoltaic (PV) Modules		Remark
Make	550Wp ,GOLDI SOLAR	Sr No.: as per attached sheet
Quantity	710 Nos	
Capacity	390.5kWp	
Inverter		Remark
Make	GROWATT	SN:-1) ADNME1900F 2)LELEE1505J 3)LELEE1506X
Quantity	3 NOS	
Capacity	125KW-2 Nos. & 50 KW-1 Nos	
	300 KW	

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As clarified earlier, there is no mention of 0.4MW solar capacity in any of NWMPL documents.

About the mismatch of the Consumer No. in the Work Completion report, we took up this matter to the concerned MSEDCL Officer, who informed NWMPL that the incorrect Consumer No is due to typographic error and by mistake this Consumer No 049084771715 has been typed in place of 049139016010.

As discussed, with the KBS, an email request will be sent to MSEDCL office, however this will have to be sent from NMC side as the Consumer No is on the Name of Commissioner Nashik Municipal Corporation. Thus, NWMPL will request NMC to send the mail to MSEDCL from their side.

Documentation provided by Project Owner	
Work Completion Report is in the drive, folder 2: Single Line Diagram of Solar Power Generation Plant	
UCR Project Verifier assessment	**Date:** 09/10/2025

PP has explained that a 390 kW DC system was installed to support a 300 kW AC inverter capacity, accounting for conversion losses. This explanation is technically acceptable, aligns with standard solar PV design practices, and is supported by the Work Completion Report as reviewed by the verification team.

PP clarified that it resulted from a typographical error in the MSEDCL Work Completion Report, where the consumer number 049084771715 was incorrectly mentioned regarding the consumer number mismatch. Further, the verification team cross-checked the same from the electricity bills generated for the monitoring period and confirms that 049139016010 is the actual consumer number. Further, PP has submitted a letter issued by NWMPL to NMC dated 28/10/2025, titled “Correction of the NMC Consumer No in the MSEDCL Work Completion Report,” referencing letter no. AEE/Dwarka/TYR/2023-24/3407 dated 26/07/2024, requesting NMC to forward the correction request to MSEDCL. Thus, the verification team confirms that the justification is accepted. **Hence, CL 02 is closed.**

CL ID	03	**Section no.**		**Date:** 19/08/2025
Description of CL				

PP is requested to clarify the following and provide appropriate supporting documents, including the relevant calibration certificates and electricity generation bills/invoices:

1. During the on-site audit, the verification team observed the electricity meter with serial number 24005723; however, calibration certificates have been provided for meters with serial numbers 24005723 and 24051021. Additionally, the “Electronic Trivector Meter Routine Test Report” dated 18/04/2024 for meter 24005723 indicates a calibration due date of 30/05/2024, whereas the meter was observed to be installed on 13/09/2024. PP is requested to provide details of the old meter that was replaced and further explain the role of meter 24051021.

Project Owner’s response	**Date:** 29/08/2025

1. Please refer to CL 1.1, which explains the differences in the various metres mentioned in these

Assessment Findings by KBS, and, each meter's scope. The document has been attached again below.

ABT NET METER	SR NO-24005723	A solar net meter is a bi-directional meter that measures both the electricity your solar panels send to the grid and the electricity you import from the grid. A SOLAR ABT NET METER refers to this type of bi-directional net meter that also incorporates or operates within an <u>Availability-Based Tariff (ABT) system</u>. In essence, the net meter tracks your solar energy contributions and consumption, while the "ABT" designation indicates it's used in a system where electricity pricing and charging are based on the real-time availability of power supply.	
GENERATION METER	SR NO-X2246958	This meter is installed to measure the gross amount of electricity generated by your solar PV system.	
ACDB MFM METER	SR NO-X2318735	In the context of solar energy, "MFM" most commonly refers to a Multi-Function Meter, a sophisticated digital device for measuring and monitoring various electrical parameters like voltage, current, power, and energy, essential for tracking solar system performance and managing energy consumption	

Documentation provided by Project Owner

Drive – List of Documents - Folder 19. Logbooks: database for each monitored parameter

UCR Project Verifier assessment | **Date:** 24/09/2025

1. PP to provide a correct and appropriate response, including the relevance of meter 24051021. In addition, the PP to explain why meter 24005723 has a calibration due date of 30/05/2024, as previously raised. It is also noted that the calibration report for meter 24051021 is still located in the folder titled "25. Electricity Generation and Consumption Logs." **Hence, CL 03 is open.**

Project Owner’s response	Date: DD/MM/YYYY
CL03. As per the manufacturer the Calibration has to be carried out before 30/05/2024 and this meter after purchase by NWMPL was handed over to MSEDCL Authority for testing and calibration, as MSEDCL is the competent authority to test it and certify it. Accordingly, MSEDCL has carried out the testing of this device and the Test Report dated 18th April 2024 of its Calibration in the appropriate Testing Centre (Certificate issued by Schneider Electric India) is attached in Folder 25: Electricity generation and consumption logs in the drive. The meter No. 24051021 is named Current Transformer which is installed in the Solar Control Unit and it has actually no role in the Electrical generation but is one of the accessories of the Solar unit.	
Documentation provided by Project Owner	
Drive - Folder 25:Electricity generation and consumption logs	
UCR Project Verifier assessment	**Date:** DD/MM/YYYY
CL03 PP clarified that meter 24005723 was calibrated by MSEDCL on 18/04/2024, prior to the due date of 30/05/2024. Further, meter 24051021 was identified as a current transformer with no role in generation measurement, which has been verified and found acceptable by the verification team. **Hence, CL 03.1 is closed.**	

CL ID	04	Section no.		Date: 19/08/2025
Description of CL				
1. During the on-site audit, the verification team noted that no training records are available prior to 2020. Although an annual training plan exists, it was not followed. For instance, only one fire safety training and one first aid training were conducted in 2019. PP is requested to explain the deviation from the training plan and provide supporting documents, including evidence of trainings conducted during the monitoring period. 2. During the on-site audit, the verification team observed that although safety training is provided to rag pickers, they were not wearing any safety equipment. PP to clarify the same.				
Project Owner’s response				**Date:** 29/08/2025
1. Safety Training Record Availability The Site operations safety instructions and measures were being practiced regularly at NWMPL site from Feb 2017 onwards. However, in the early stages of the project, there was less documentation done, thus, concrete documents of such trainings are unavailable at the site. Although Safety Trainings and Plan were in position from 2017 onwards, the proper and systematic documentation is available only from mid-2019 onwards. Also, due to an old office computer system damage, previous data and documents lost before 2019 period could not be retrieved for records. 2. As displayed in document no. 24. Safety Training Records in the drive, the Rag Pickers have been provided with all the safety equipment, however the rag pickers continue to not wear the equipment regularly. More stricter implementations of these safety standard are being undertaken presently.				
Documentation provided by Project Owner				
UCR Project Verifier assessment				**Date:** 24/09/2025

1. PP has acknowledged the absence of training records prior to 2020, citing early project-stage limitations and data loss due to system damage. While limited evidence is available for trainings conducted before mid-2019, the PP has demonstrated that a structured training plan and systematic documentation have been in place since then, and the same is accepted by the verification team based on the justification and supporting documents provided. **Hence, CL 04.1 is closed.**
2. PP has confirmed that safety equipment was provided to rag pickers but acknowledged inconsistent usage, as observed during the on-site visit. Stricter enforcement measures are now being implemented. Based on the clarification and supporting documents, the verification team finds the response acceptable. **Hence, CL 04.2 is closed.**

CL ID	05	**Section no.**		**Date:** 19/08/2025

Description of CL

PP to provide the following documents, as the same is not accessible/irrelevant/provided:

1. Document is irrelevant for Contracts between local contractors/kabari, thereby providing monetary benefits to the local population, as PP provided the “CSR activities for RAG pickers”.
2. PP to submit the complete temperature sensor logs for the monitoring period and update the MR accordingly.
3. Supporting documents not provided in the folder “Any revenue records from compost, energy or plastic fuel sales”.
4. OHS compliance records provided are outside the monitoring period and not relevant; PP to submit supporting documents applicable to the current monitoring period.
5. Environmental clearance

Project Owner’s response	**Date:** 29/08/2025

1. Free provision of dry waste to the rag pickers, who in turn sell it in the Indian recycling market for additional income. Diwali presents are distributed during Diwali. Regular health checkups are done for the ragpickers. The NGO that organizes the ragpickers does not maintain records of the health check-ups and Diwali gifts.
2. Temperature monitoring of the Organic Waste Composting Process

   The organic waste component from the MSW is treated by an Aerobic Composting process. This process involves aerobic decomposition of the bio-degradable waste at a temperature range between 35 degrees to 65 degrees Celsius in different processing stages.

   The temperature of the composting material is being checked at specific intervals with the help of portable temperature probe for all the compost batches and the respective temperatures displayed on the Portable Thermometer device are recorded and noted manually. Based on the temperature of the process material, the process time and stages are accordingly maintained.

   We have uploaded the photographs of the temperature measurement activity for easy understanding in the drive, folder 28. Temperature sensor logs.

   From the statutory point of view, it is not required to maintain the temperature noting and records for the Process Control operations of Compost Process. Therefore, the physical register or excel sheet records were maintained by the site operations team.

   But if it is required in future, we have already started maintaining the records from the current year onwards.
3. There is no financial additionality on the UCR CoU platform. Hence, there is no requirement for such evidence. This had been mutually agreed upon by both the KBS on site audit team and NWMPL team.
4. The relevant OHS compliance records are uploaded in the drive, folder 16. OHS compliance records.
5. As per the MPCB standards, the project does not require environmental clearance now as the Nashik Municipal Corporation (NMC) had already got the clearance in 2004 while setting up the project.

Documentation provided by Project Owner

UCR Project Verifier assessment	**Date:** 24/09/2025
1. The response provided by the PP is now accepted by the verification team based on the CSR-related documents submitted by the PP. **Hence, CL 05.1 is closed.** 2. PP to clarify where it is mentioned that "it is not required to record", and they have records, which are maintained by site operational. **Hence, CL 05.2. is open.** 3. As additionality is not determined under the UCR CoU platform, and based on findings from on-site interviews, the verification team accepts this. **Hence, CL 05.3 is closed.** 4. PP has now provided OHS records from 2021 to 2025, covering one month for each year; however, data from 2017 has not been provided accurately. **Hence, CL 05.4 is open.** 5. PP to provide the clearance documents approved in 2004. **Hence, CL 05.5 is open.**	
Project Owner's response	**Date:** 01/10/2025
CL5.2. The maintaining of temperature records is not a criterion for quality control by the statutory bodies for the waste management process. It is required just to monitor the temperature changes. Accordingly, the site process team monitors the temperature changes and carries out the Process Control instead of maintaining the temperature reading records. **CL5.4.** The Site operations safety instructions and measures were being practiced regularly at NWMPL site from Feb 2017 onwards. However, in the early stages of the project, there was less documentation done, thus, concrete documents of such trainings are unavailable at the site. Although Safety Trainings and Plan were in position from 2017 onwards, the proper and systematic documentation is available only from mid-2019 onwards. Also, due to an old office computer system damage and due to the Files getting soaked because of water seepage, previous data/documents lost before 2019-20 period could not be retrieved for records. **CL5.5.** NWMPL will check for the availability of such document taken by Nashik Municipal Corporation and have accordingly informed NMC to provide us the Environmental Clearance document issued for setting up the MSW Processing Plant.	
Documentation provided by Project Owner	
EC certificate for MSW processing and disposal facility to NWMPL	
UCR Project Verifier assessment	**Date:** 12/05/2026
CL5.2. PP clarified that while maintaining temperature records is not a statutory requirement, the site team monitors temperature using a portable probe and manages the process accordingly, and the same is accepted by the verification team. **Hence, CL 05.2 is closed.** **CL5.4.** PP clarified that although OHS measures and training were implemented since 2017, documentation is only available from mid-2019 due to limited record-keeping in the initial years and data loss caused by system damage and water seepage and accepted by the verification team. **Hence, CL 05.4 is closed.** **CL5.5.** PP clarified that Environmental Clearance was not required since the project involves waste management activities. As the originally issued Environmental Clearance document is unavailable, PP has requested NMC to reissue the same. NWMPL issued a letter dt. 29/10/2025 for requesting the copy of Environmental Clearance, based on the request Maharashtra Pollution Control Board (MPCB) issued the EC certificate for MSW processing and disposal facility to NWMPL on 25/03/2026. **Hence, CL 05.5 is closed.**	

Table 2. CARs from this Project Verification

CAR ID	01	**Section no.**		**Date:** 19/08/2025
Description of CAR				
PP to explain the inconsistency in the project start date, which is stated as 15/02/2017 in the MR and PCN, while the supporting documents indicate that the project has been operational since 15/01/2017.				
Project Owner's response				**Date:** 29/08/2025

The physical activities at site had started from 15 Jan 2017 and the first invoice was raised on 15th Feb 2017 --- so NMC audit team considered the start of this project w.e.f. the first billing date.
Documentation provided by Project Owner
Drive List of Documents: Folder 1. Evidence for start date of project activity

UCR Project Verifier assessment	**Date:** 24/09/2025
PP has clarified that physical project activities commenced on 15/01/2017, while the first invoice was issued on 15/02/2017, which was previously considered the start date based on the first billing. **Hence, CAR 01 is closed.**	

CAR ID	02	**Section no.**		**Date:** 19/08/2025
Description of CAR				
PP has provided one salary slip from each year covering the monitoring period; however, the total number of people employed with NWMPL during this period, including long-term, short-term, permanent, and temporary roles, is unclear. The number of women employees is also not clearly indicated. PP to provide detailed employment information along with supporting documents.				
Project Owner's response				**Date:** 29/08/2025
The detailed number of employees are provided in the following table:				

Year	Total No of employees	No. of Male Employees	No. of Female Employees	No. of Permanent Employees	No. of Contract Employees
2017	33	20	1	21	12
2018	76	58	2	60	16
2019	84	64	2	66	18
2020	128	105	3	108	20
2021	114	101	1	102	12
2022	113	97	1	98	15
2023	132	101	1	102	30
2024	160	124	4	128	32

Documentation provided by Project Owner	
Drive : List of Docs_KBS_NWMPL_Audit , employment data	
UCR Project Verifier assessment	**Date:** 24/09/2025
The number of employees from 2017 to 2024 has been provided by the PP, along with the supporting documents, and the same has been reviewed and accepted by the verification team. **Hence, CAR 02 is closed.**	

CAR ID	03	**Section no.**		**Date:** 19/08/2025
Description of CAR				
As per the UCR MR template, values achieved during the monitoring period for the monitored parameters should be mentioned. However, they are not mentioned in MR. Also, the monitoring plan description should be there.				
Project Owner's response				**Date:** 29/08/2025
The monitoring plan description has been added to the MR version 1(Page 60), and is available in the drive.				
Documentation provided by Project Owner				
Drive – List of Docs_KBS_NWMPL_Audit – Monitoring Report				
UCR Project Verifier assessment				**Date:** 09/09/2025

The monitoring plan description has now been revised in the MR; however, the MR does not adequately address information regarding data storage and archiving procedures; accuracy and quality assurance measures; details of monitoring personnel; calibration and verification procedures, including calibration frequency; recording methods; training and maintenance procedures; and the organizational chart, etc. PP to revise the same in the section C.10 of the MR. **Hence, CAR 03 is open.**

Project Owner's response	**Date:** 01/10/2025
The Monitoring Plan description with the requested details has been added in the MR – C.10. Monitoring Plan.	
Documentation provided by Project Owner	
UCR Project Verifier assessment	**Date:** 09/10/2025
PP has now updated the monitoring plan in Section C.10 of the MR, which is accepted by the verification team. **Hence, CAR 03 is closed.**	

Table 3. FARs from this Project Verification

FAR ID	00	**Section no.**		**Date:** DD/MM/YYYY
Description of FAR				
NA				
Project Owner's response				**Date:** DD/MM/YYYY
NA				
Documentation provided by Project Owner				
NA				
UCR Project Verifier assessment				**Date:** DD/MM/YYYY
NA				